Lab Assessment 1: Data Cleaning and Preparation

Using Power BI (Power Query) and KNIME

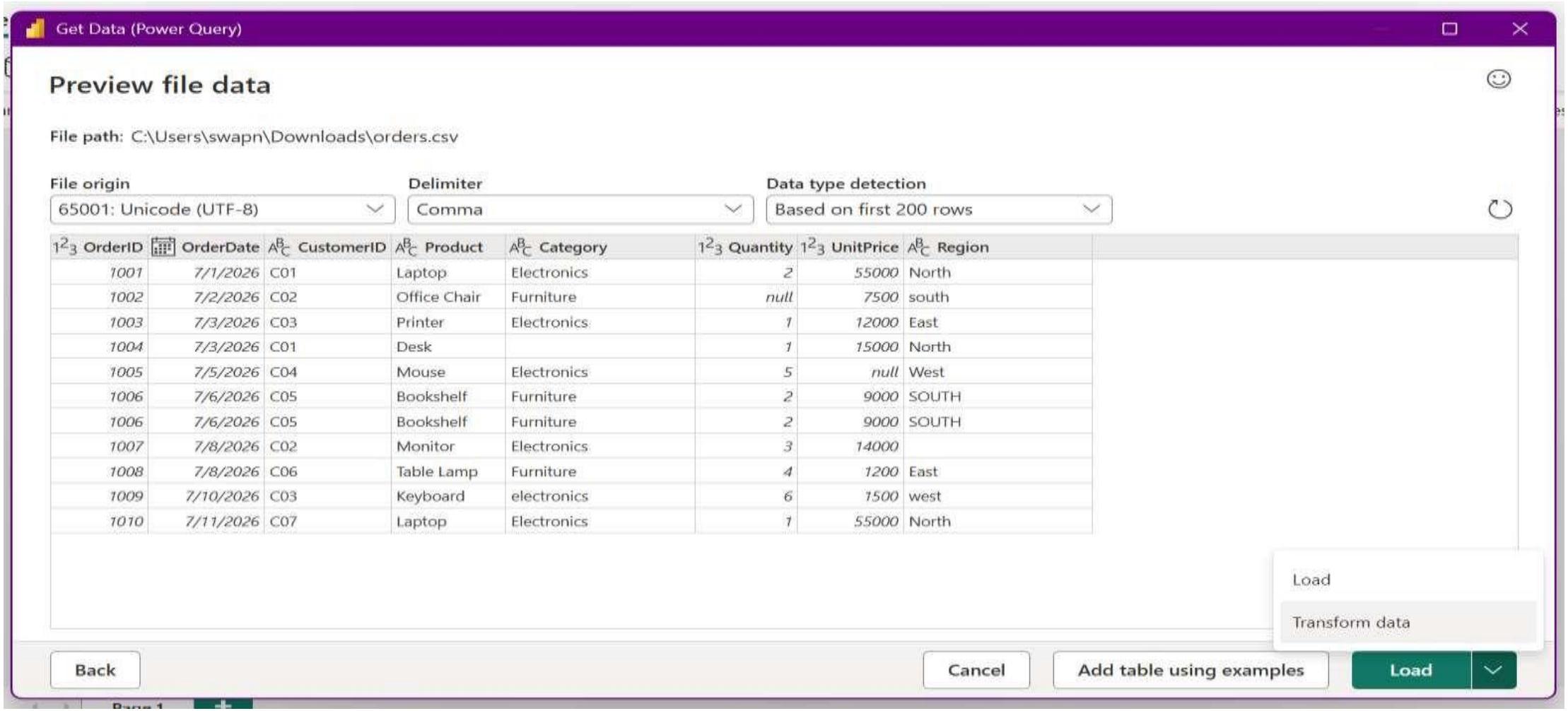
Data Analysis with Power BI & KNIME (SECSE001) | MCA Semester 1 | K.R. Mangalam University

Name	**Shivam Saini**
Roll no.	KRMU2660286

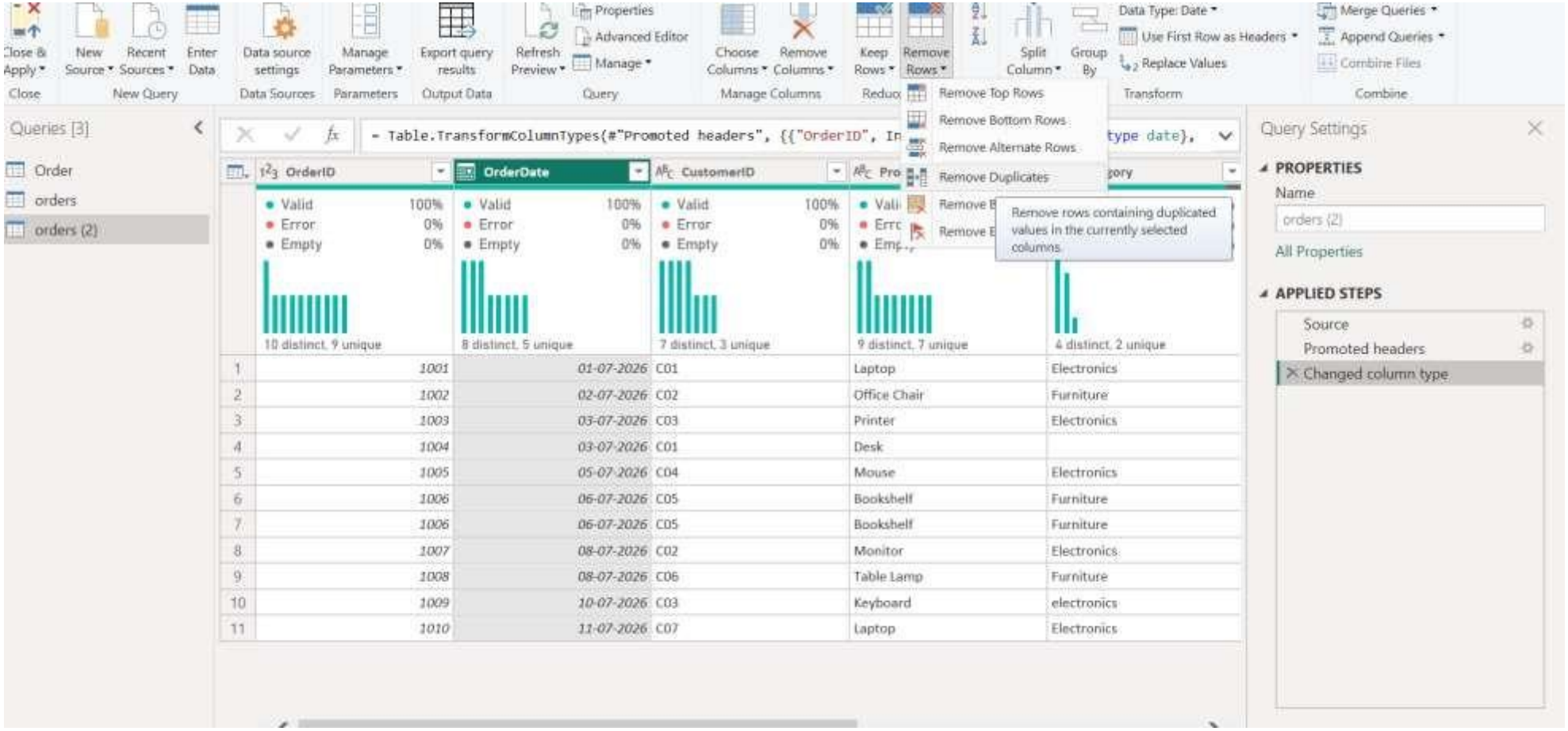
Part-1 (Data Cleaning and Preparation)

Power BI (Power Query)

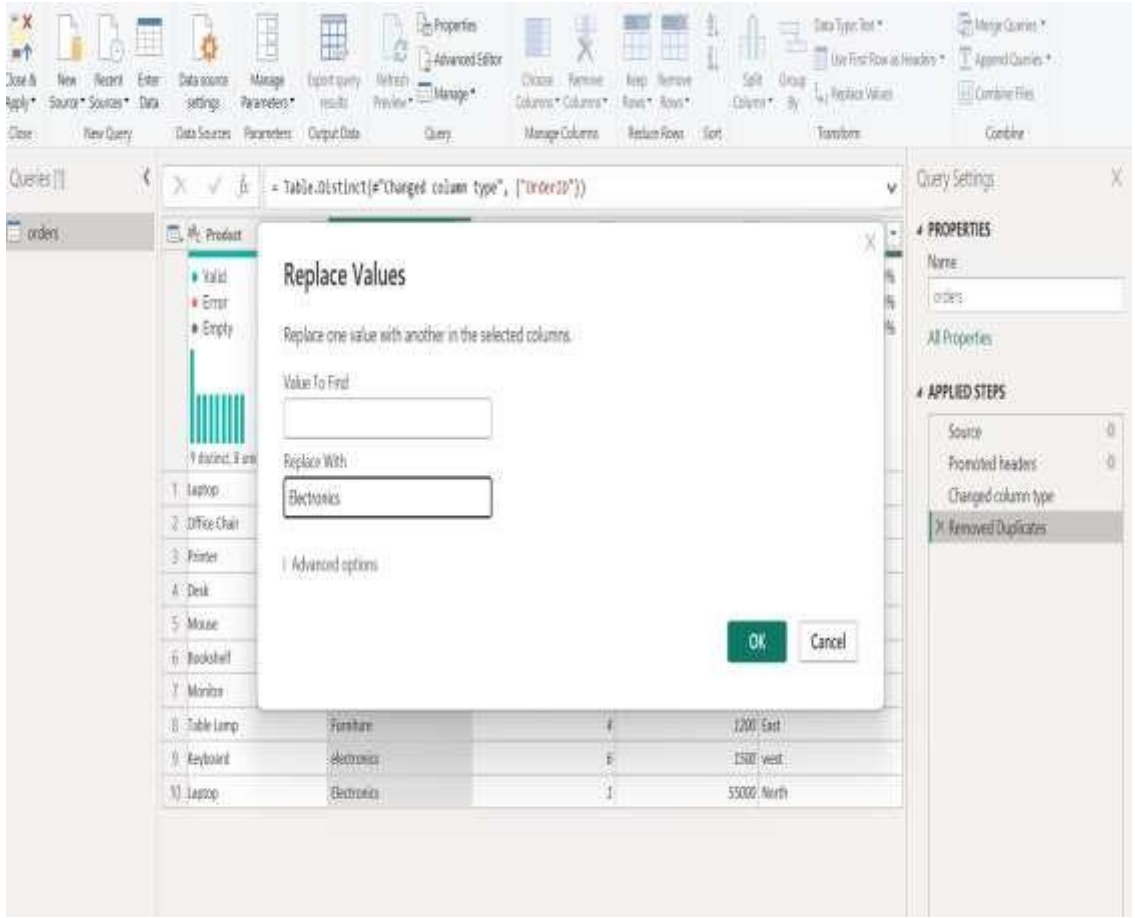
Loading Order.csv File

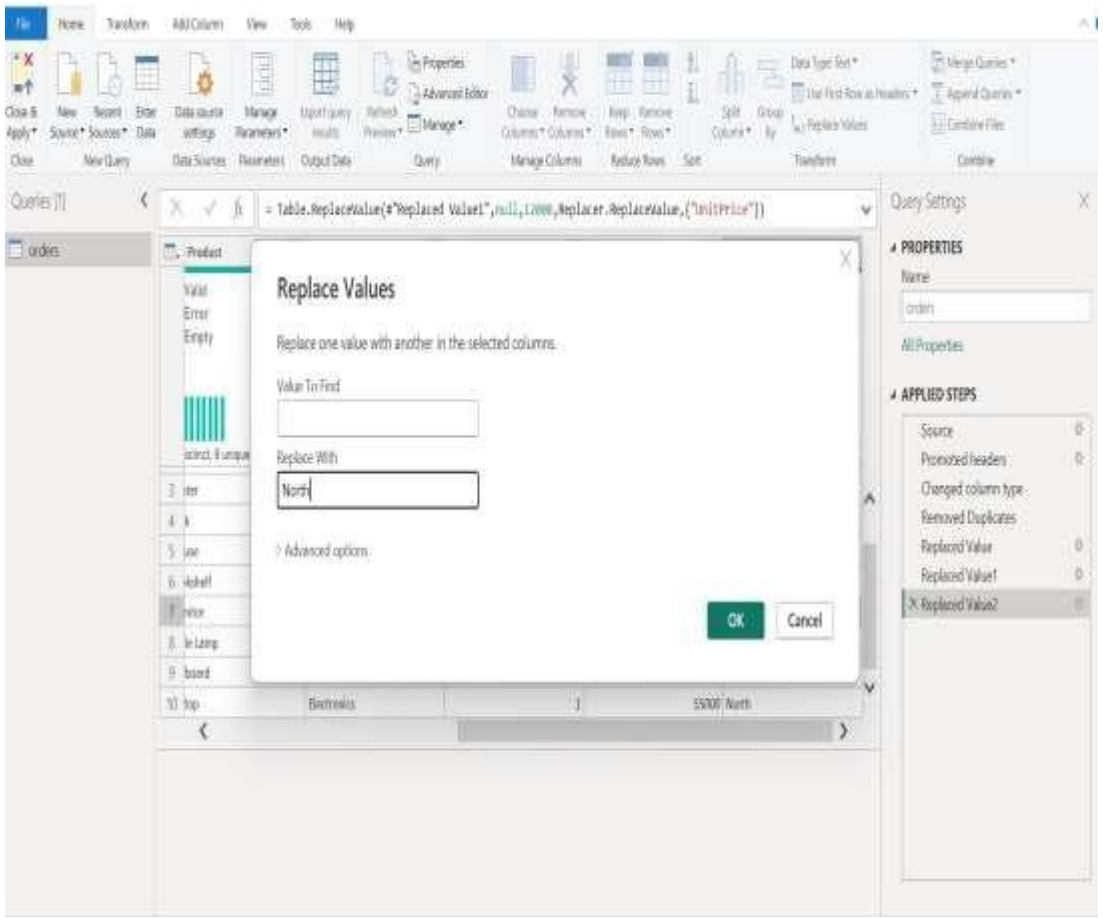


Removing Duplicate Rows

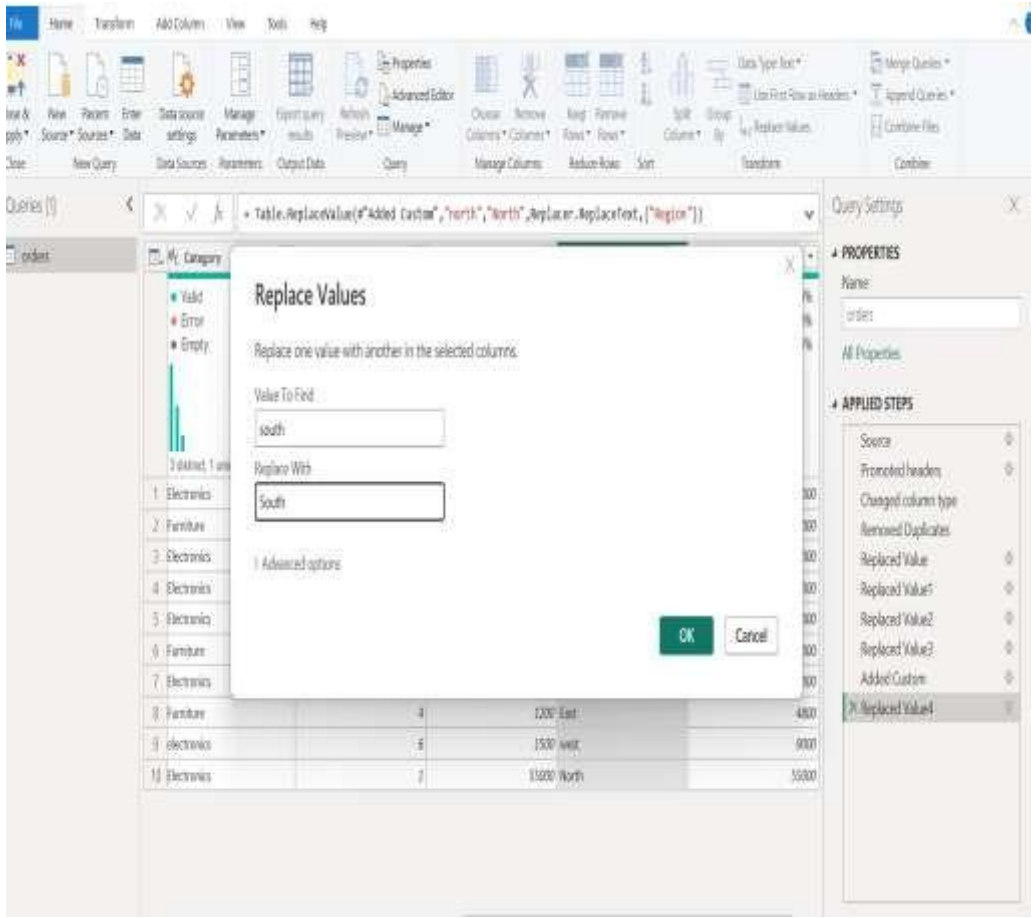


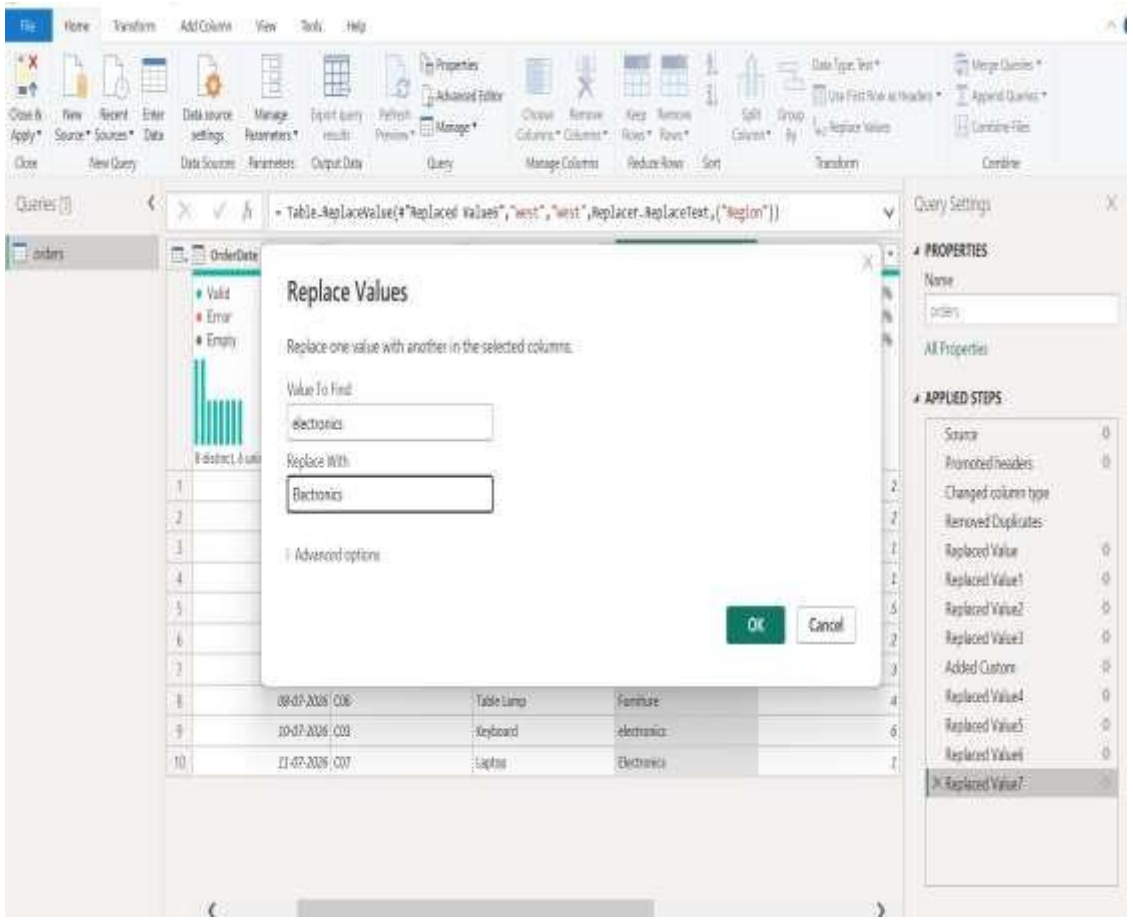
Replacing Null Values



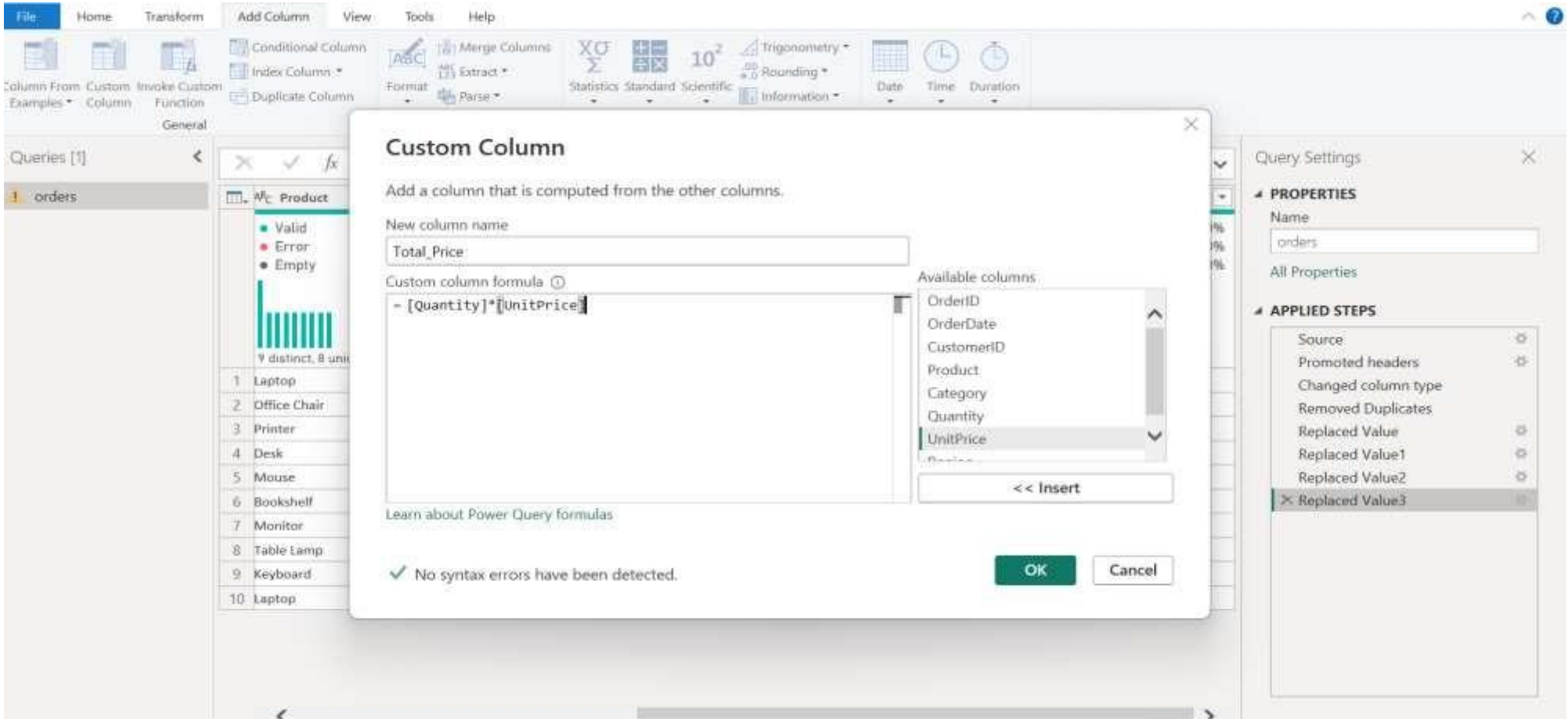


Standardized Columns

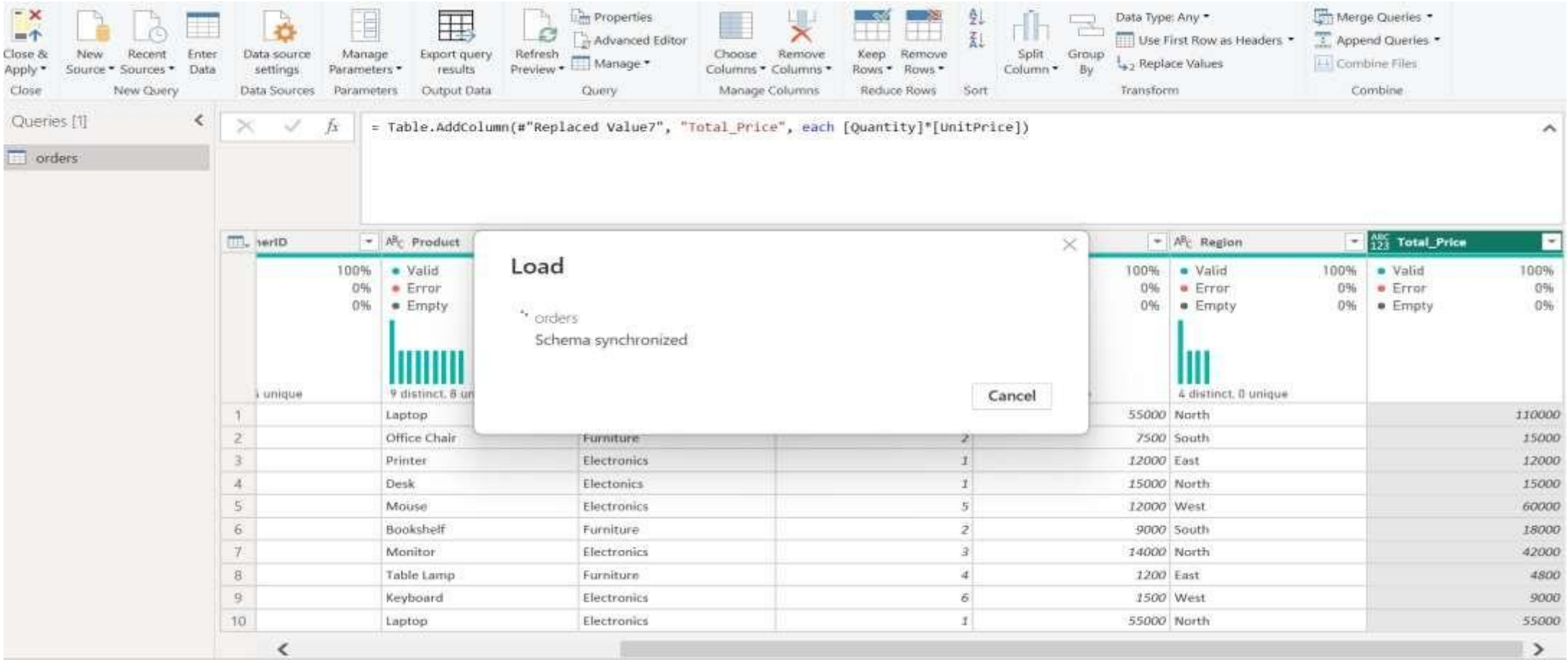




Adding Column Total_Price

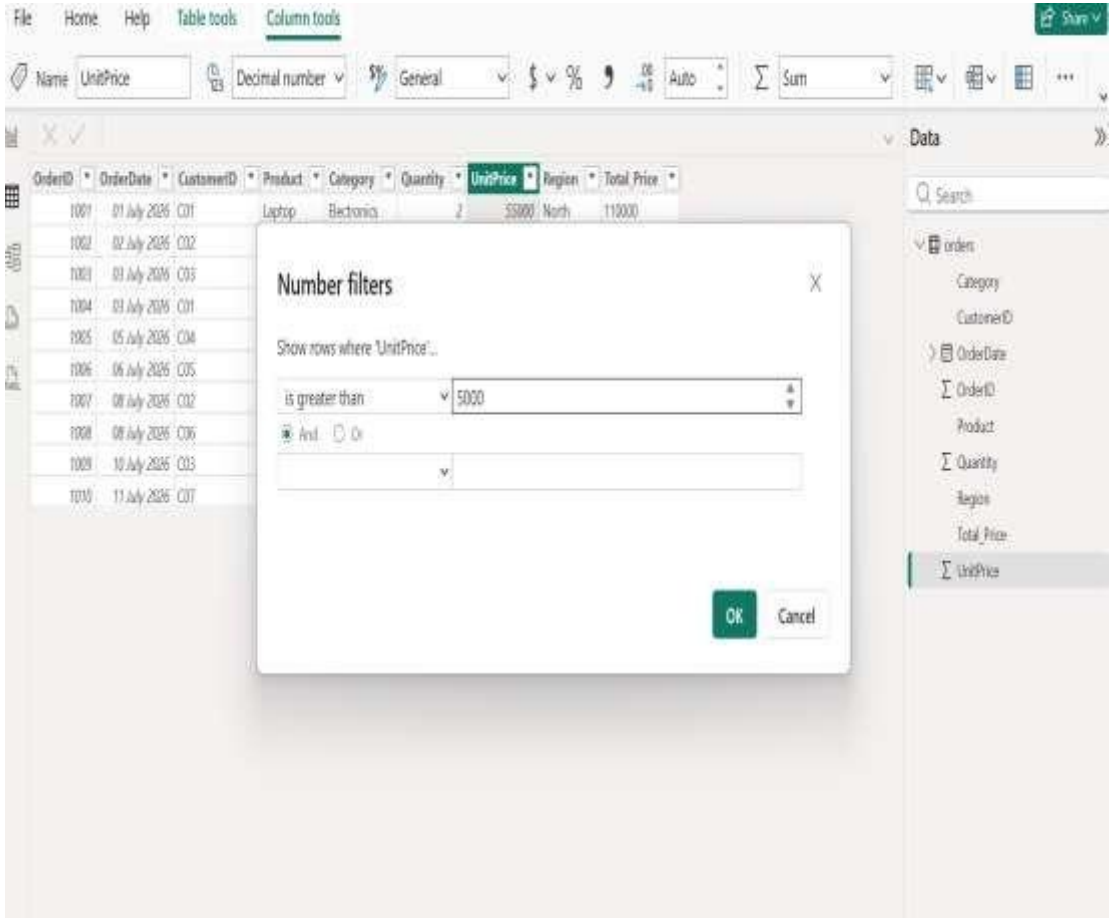


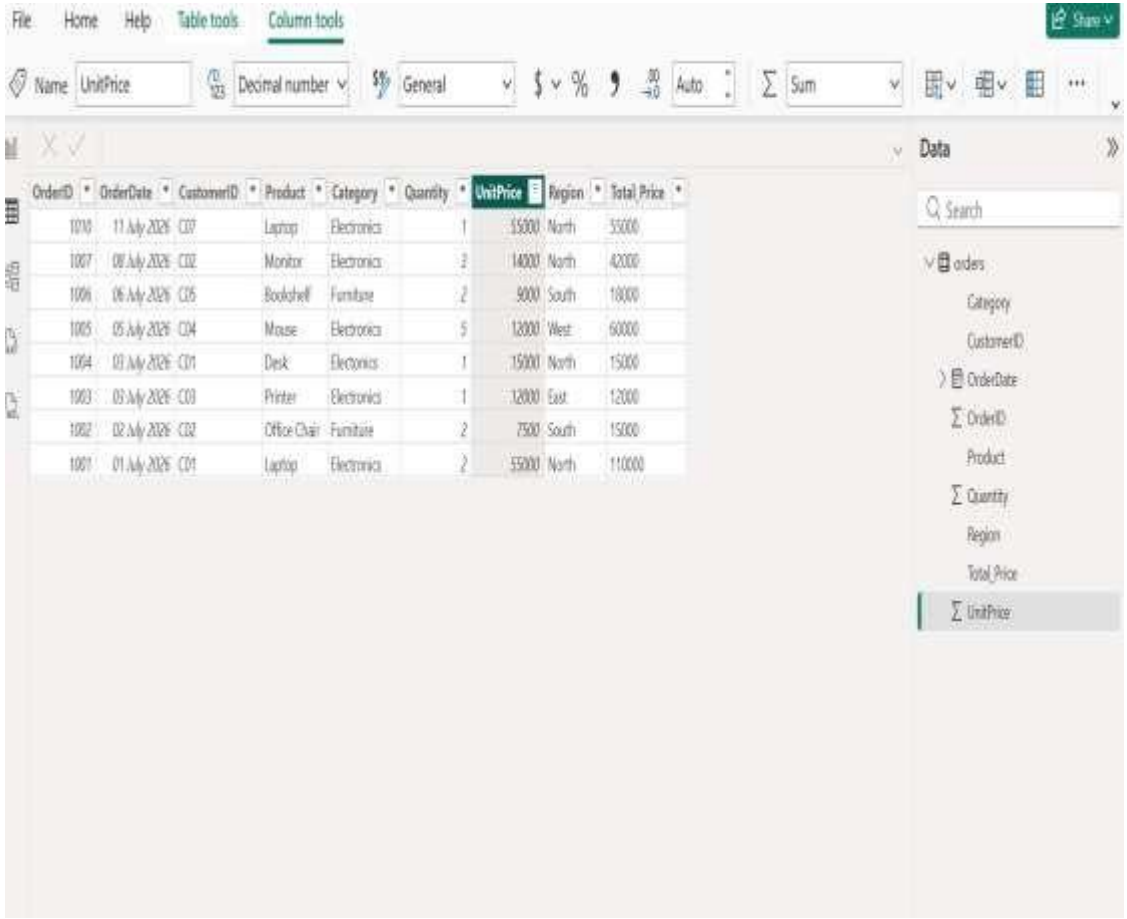
Close and Apply the File



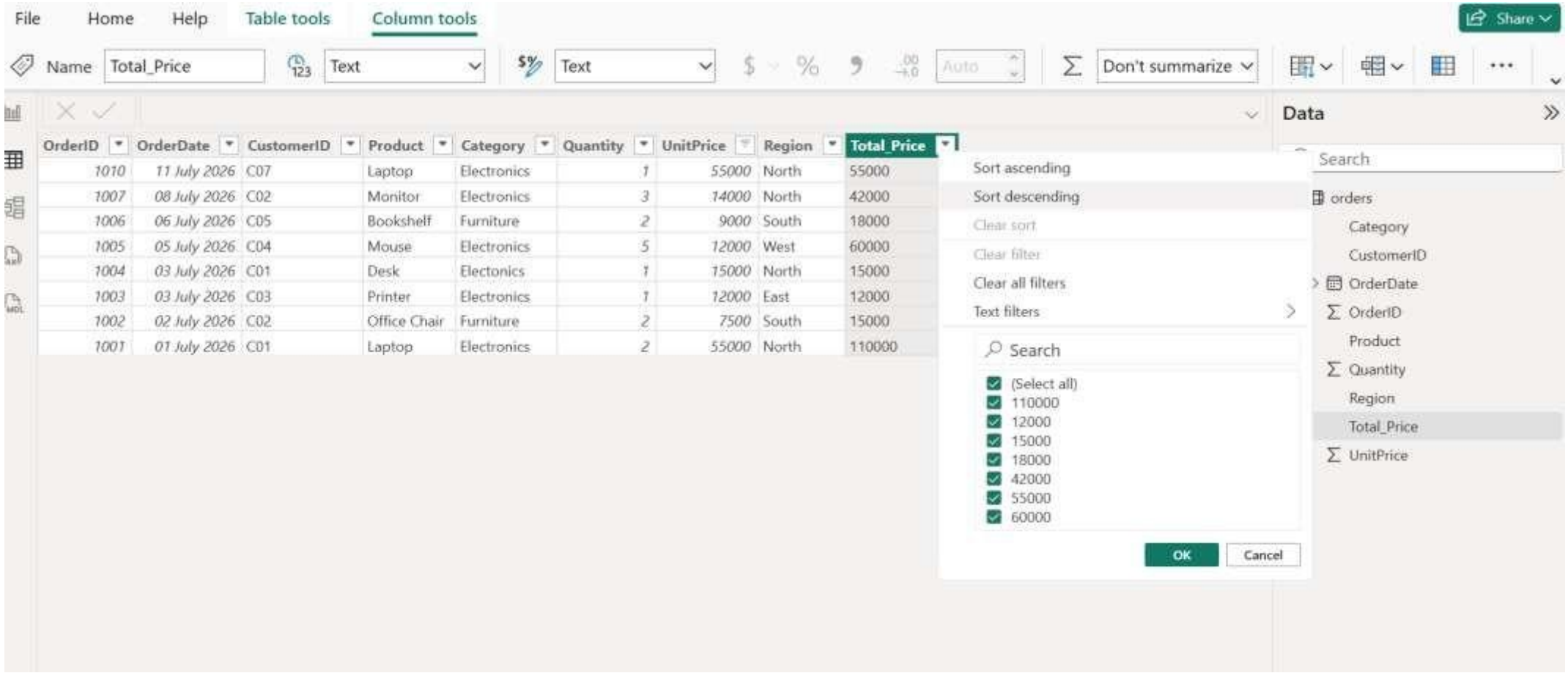
	Product				Region	Total_Price
1	Laptop			55000	North	110000
2	Office Chair	Furniture	2	7500	South	15000
3	Printer	Electronics	1	12000	East	12000
4	Desk	Electonics	1	15000	North	15000
5	Mouse	Electronics	5	12000	West	60000
6	Bookshelf	Furniture	2	9000	South	18000
7	Monitor	Electronics	3	14000	North	42000
8	Table Lamp	Furniture	4	1200	East	4800
9	Keyboard	Electronics	6	1500	West	9000
10	Laptop	Electronics	1	55000	North	55000

Filtering the rows on the basis of column



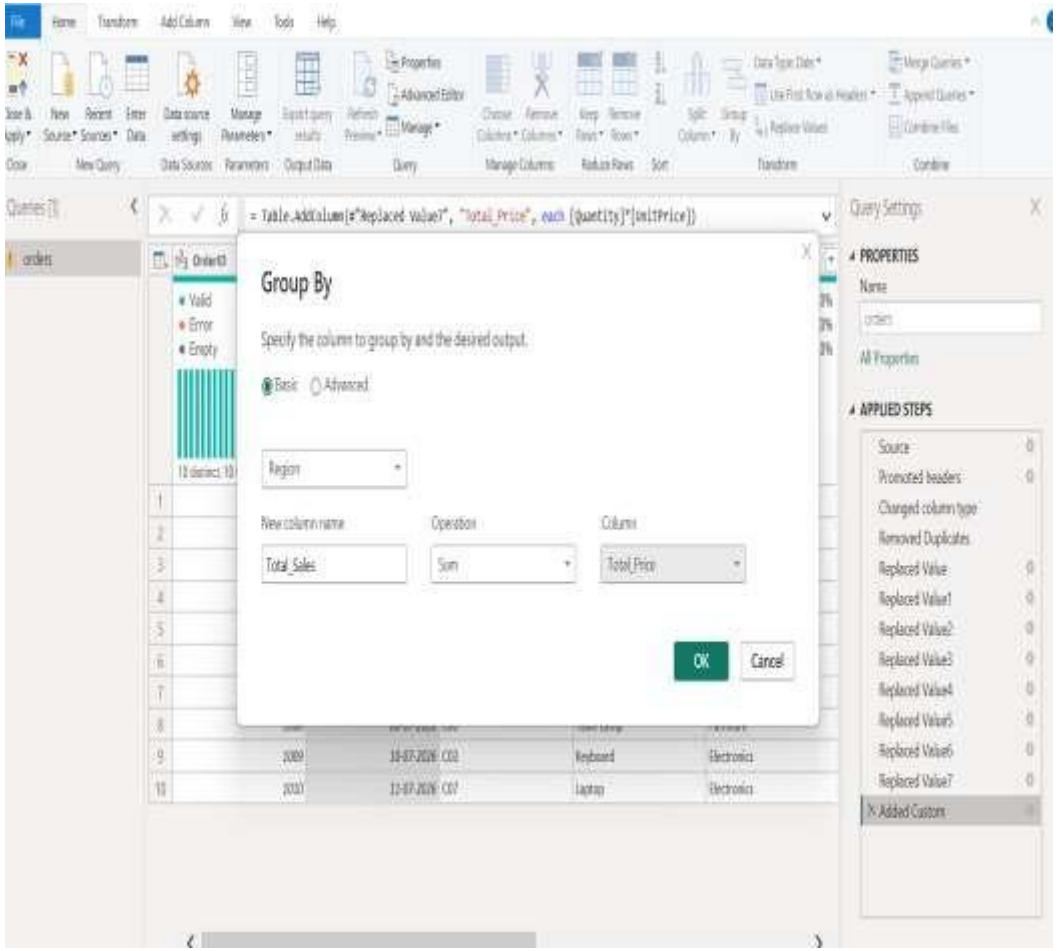


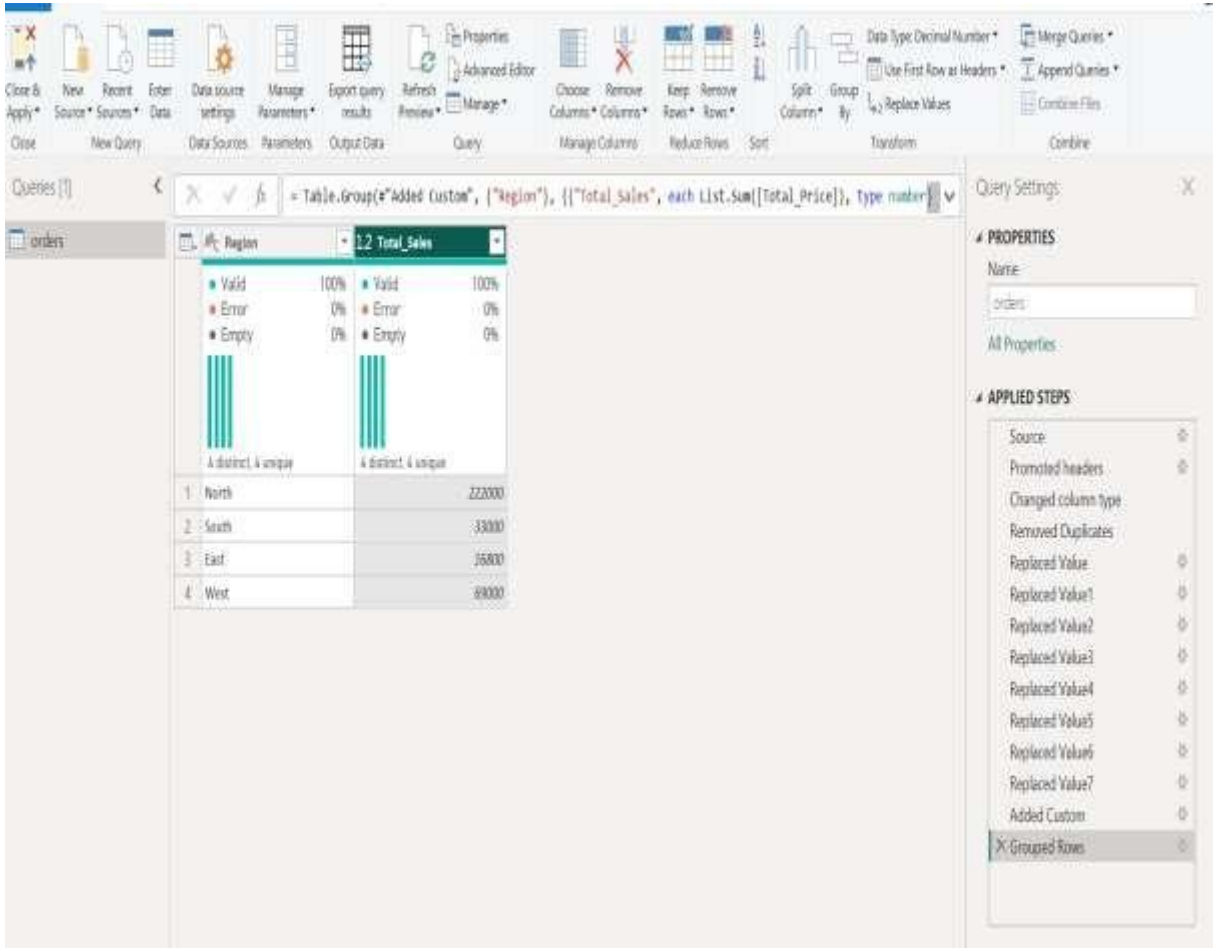
Sorting in descending order on Total_Price



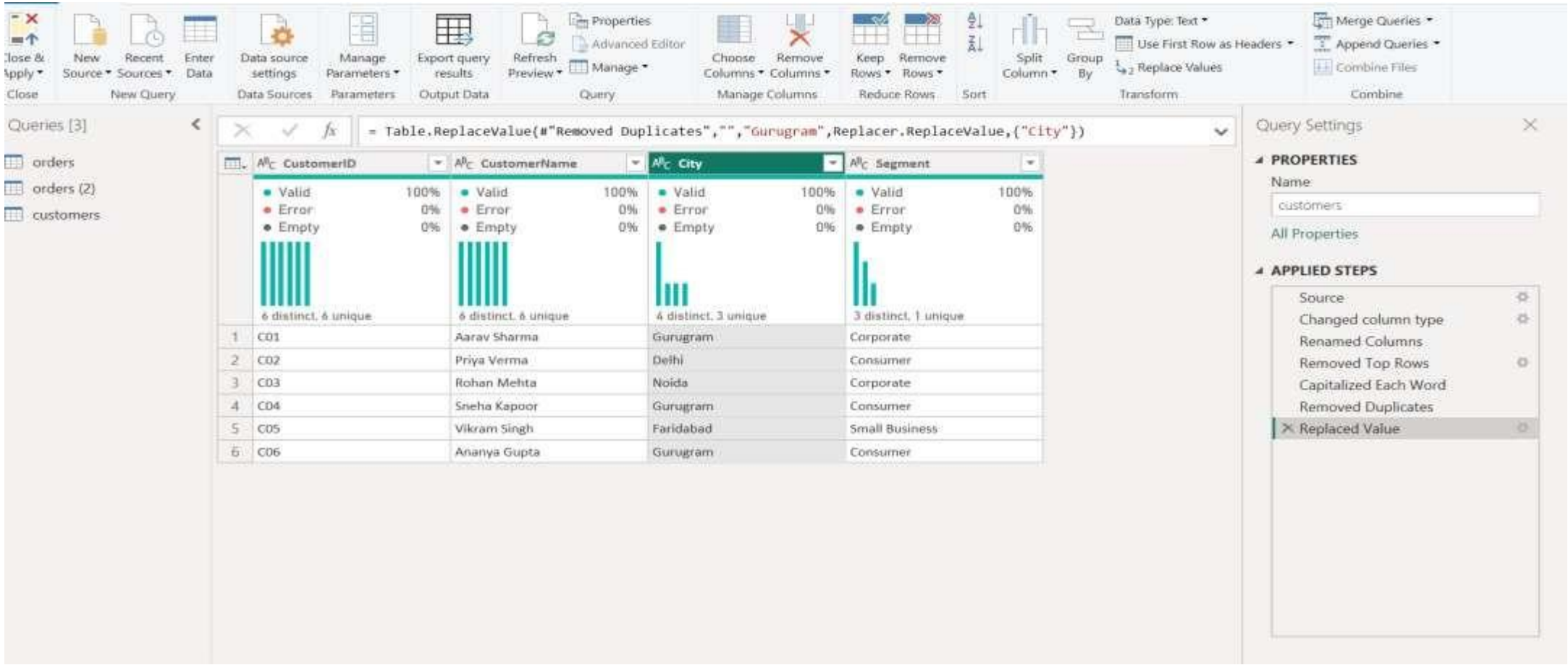
OrderID	OrderDate	CustomerID	Product	Category	Quantity	UnitPrice	Region	Total_Price
1010	11 July 2026	C07	Laptop	Electronics	1	55000	North	55000
1007	08 July 2026	C02	Monitor	Electronics	3	14000	North	42000
1006	06 July 2026	C05	Bookshelf	Furniture	2	9000	South	18000
1005	05 July 2026	C04	Mouse	Electronics	5	12000	West	60000
1004	03 July 2026	C01	Desk	Electonics	1	15000	North	15000
1003	03 July 2026	C03	Printer	Electronics	1	12000	East	12000
1002	02 July 2026	C02	Office Chair	Furniture	2	7500	South	15000
1001	01 July 2026	C01	Laptop	Electronics	2	55000	North	110000

Grouping the table on the basis region Column

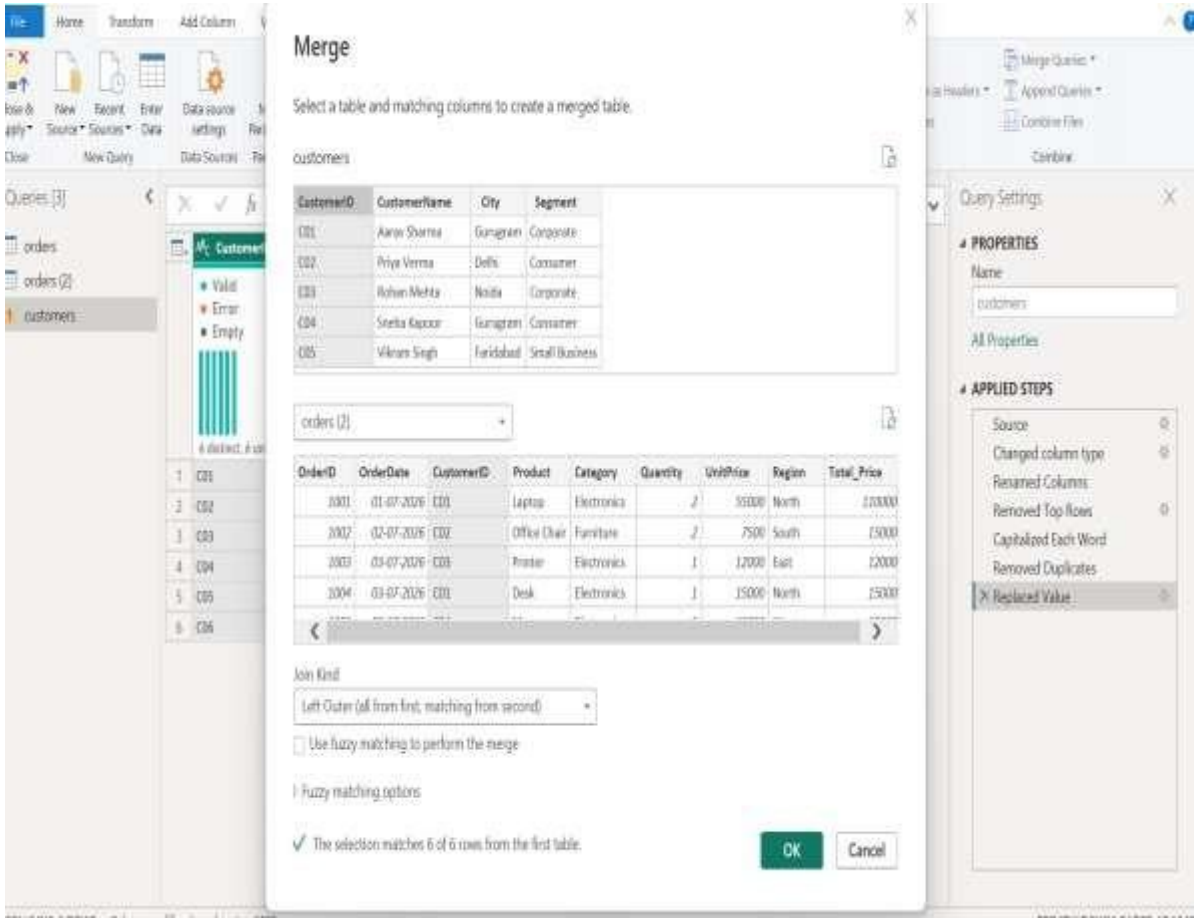


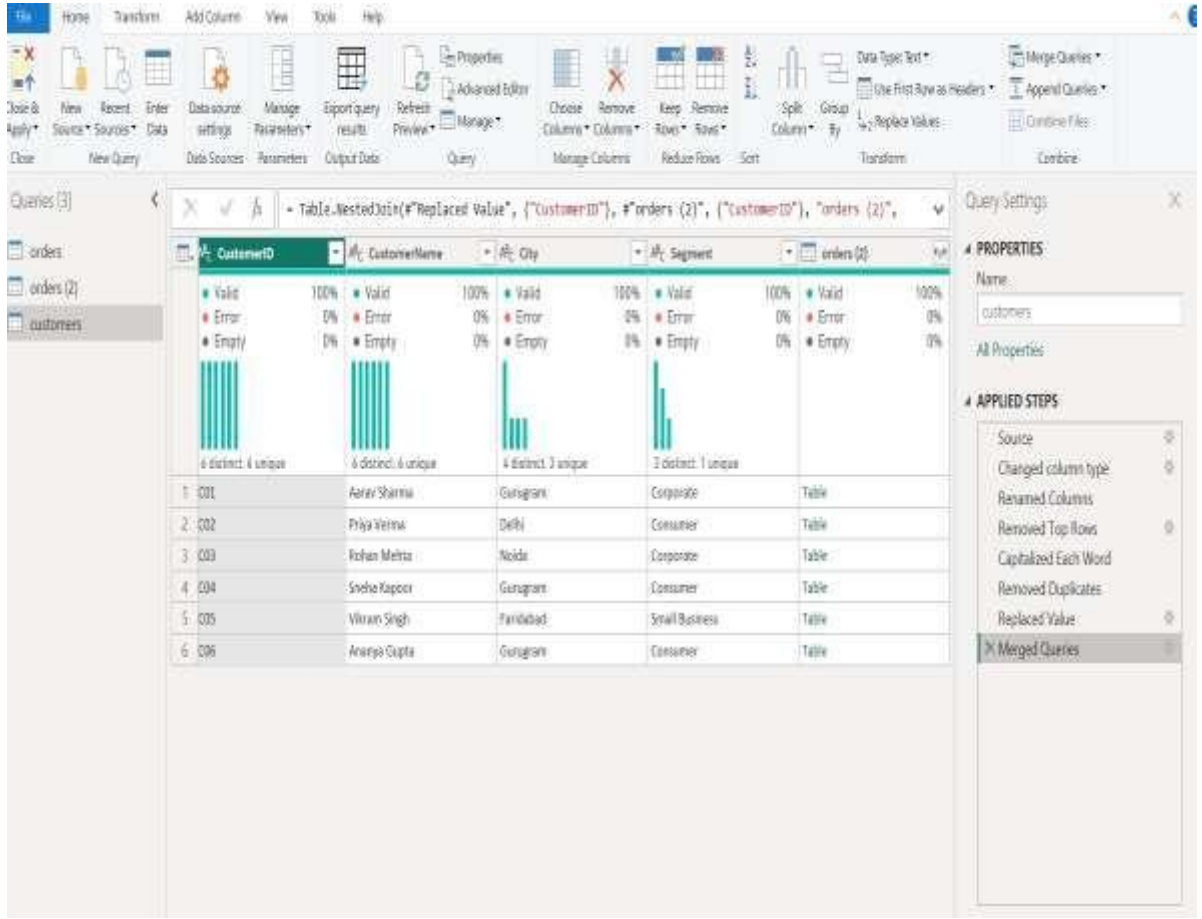


Cleaning,Standardizing on Customer.csv File

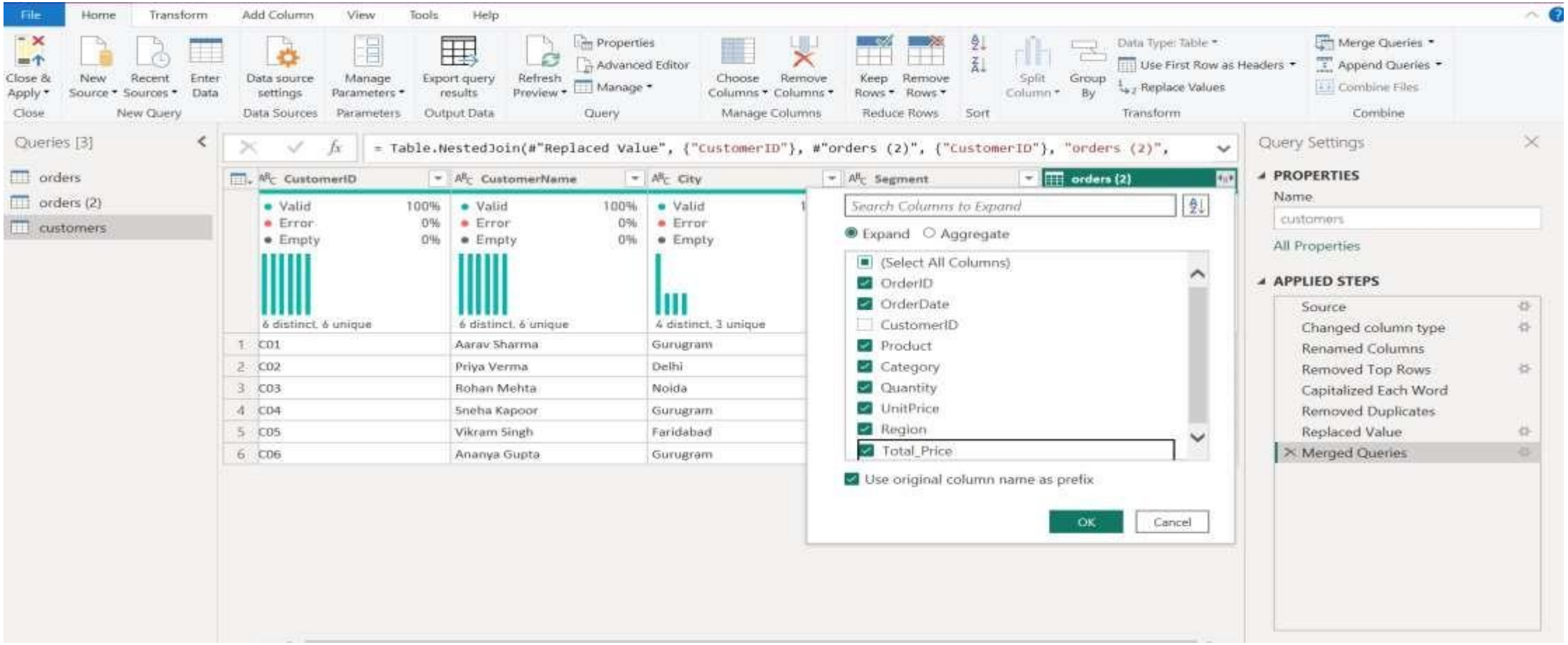


Merging Customer table to Order Table

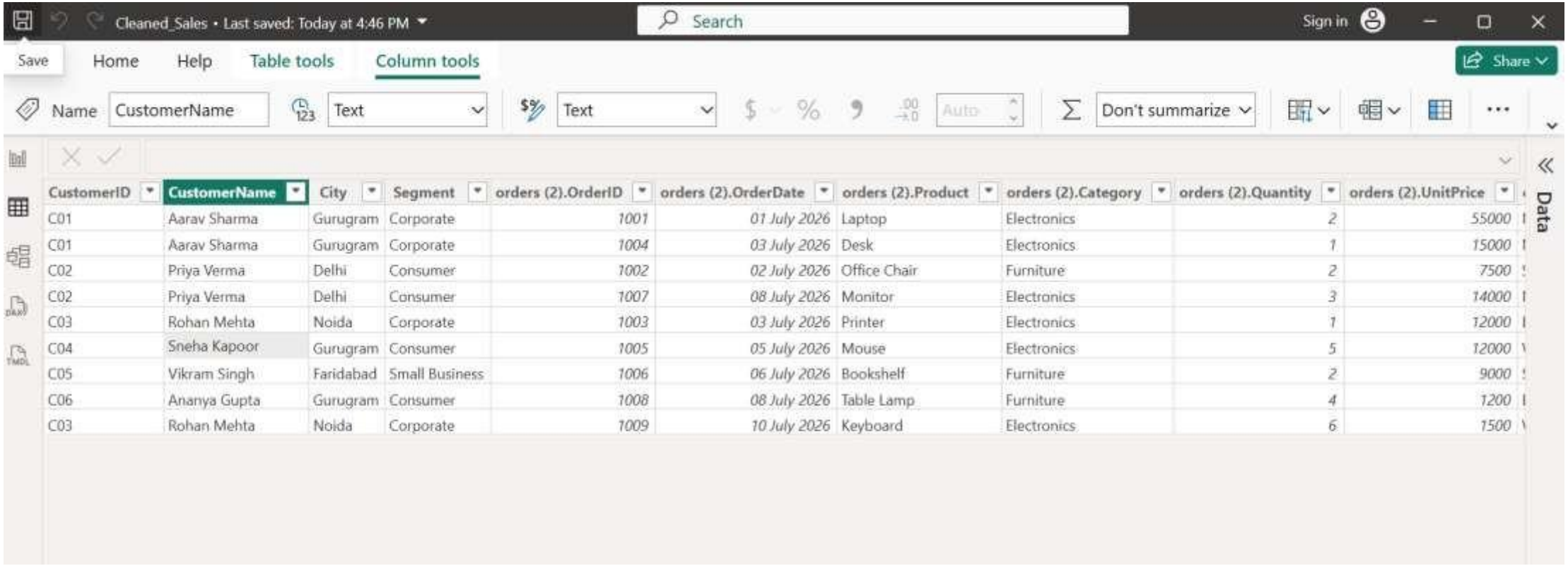




Expanding the order table after Merging



Final Cleaned_Sales.csv File

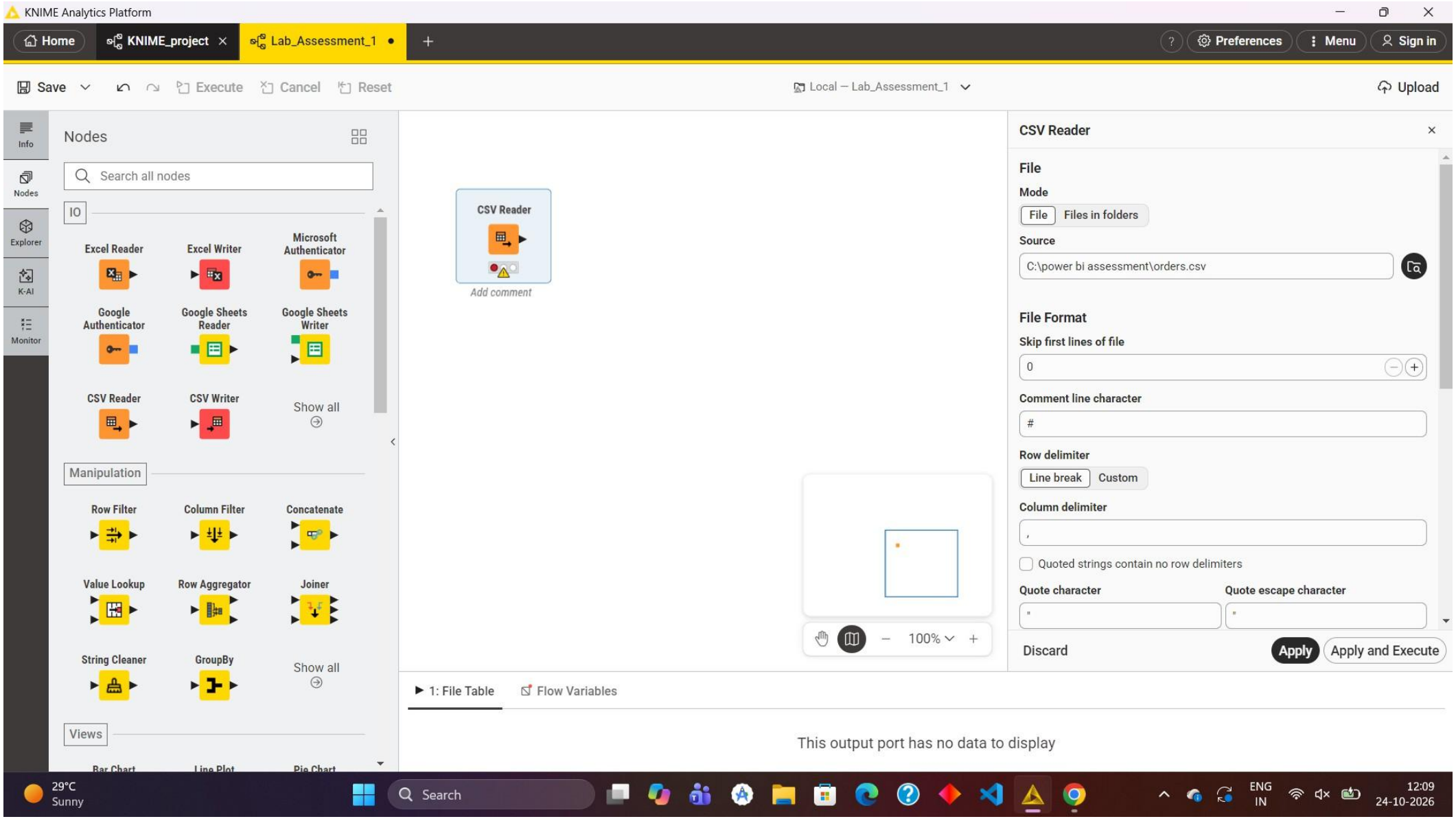


CustomerID	CustomerName	City	Segment	orders (2).OrderID	orders (2).OrderDate	orders (2).Product	orders (2).Category	orders (2).Quantity	orders (2).UnitPrice
C01	Aarav Sharma	Gurugram	Corporate	1001	01 July 2026	Laptop	Electronics	2	55000
C01	Aarav Sharma	Gurugram	Corporate	1004	03 July 2026	Desk	Electronics	1	15000
C02	Priya Verma	Delhi	Consumer	1002	02 July 2026	Office Chair	Furniture	2	7500
C02	Priya Verma	Delhi	Consumer	1007	08 July 2026	Monitor	Electronics	3	14000
C03	Rohan Mehta	Noida	Corporate	1003	03 July 2026	Printer	Electronics	1	12000
C04	Sneha Kapoor	Gurugram	Consumer	1005	05 July 2026	Mouse	Electronics	5	12000
C05	Vikram Singh	Faridabad	Small Business	1006	06 July 2026	Bookshelf	Furniture	2	9000
C06	Ananya Gupta	Gurugram	Consumer	1008	08 July 2026	Table Lamp	Furniture	4	1200
C03	Rohan Mehta	Noida	Corporate	1009	10 July 2026	Keyboard	Electronics	6	1500

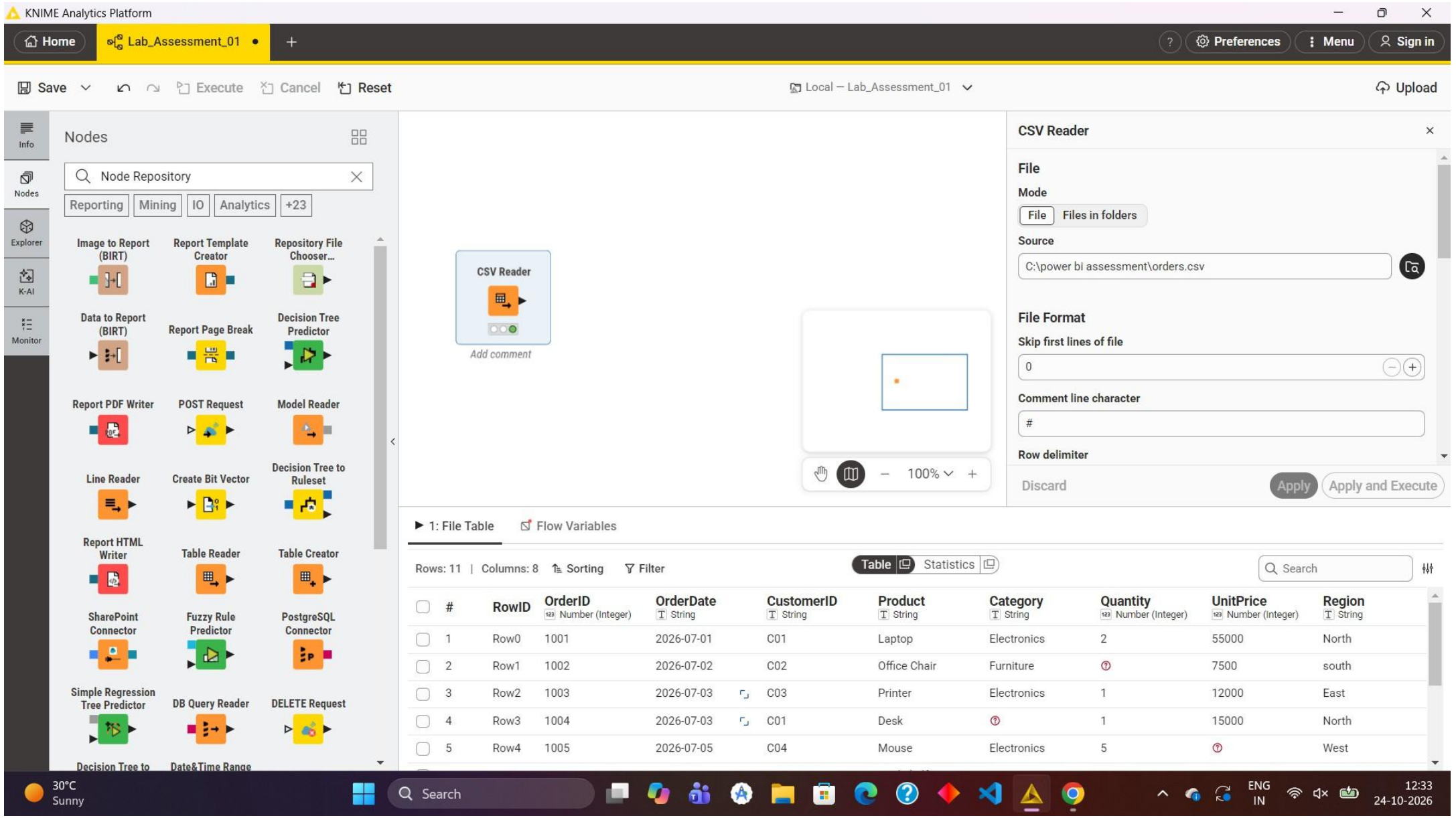
Part -2 (Data Cleaning and Preparation)

KNIME

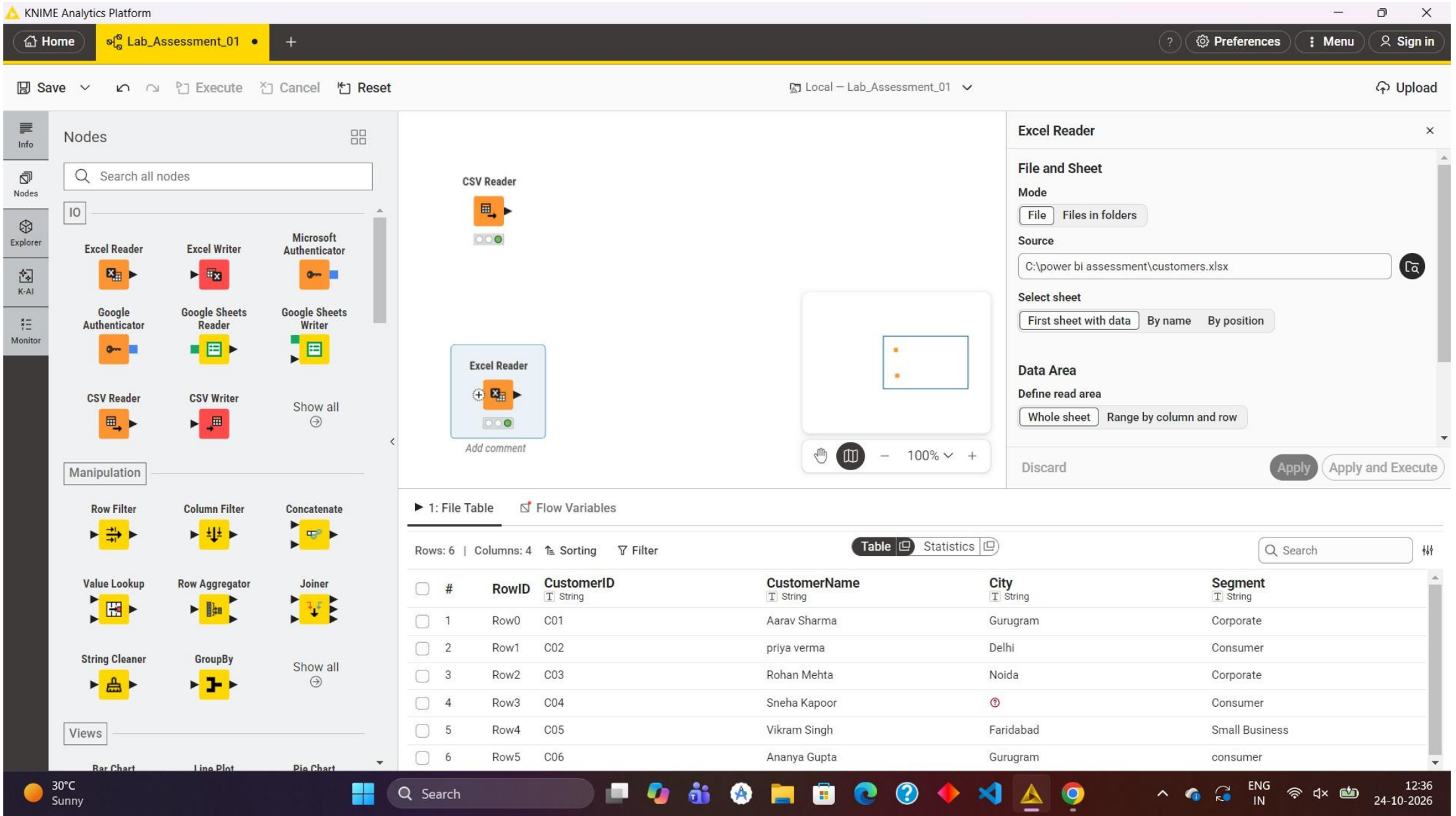
Importing the Orders Dataset



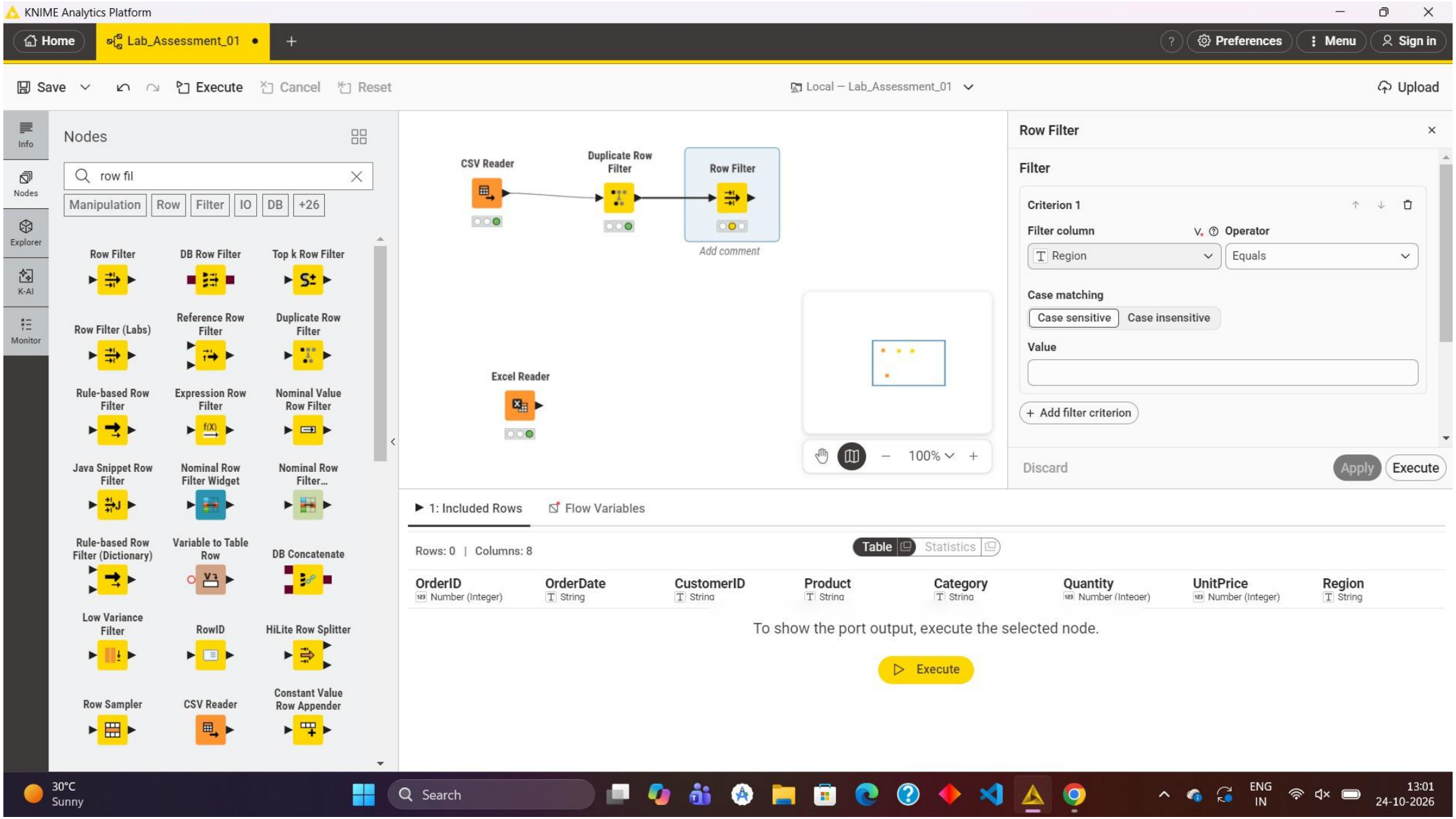
Verifying the Orders Table



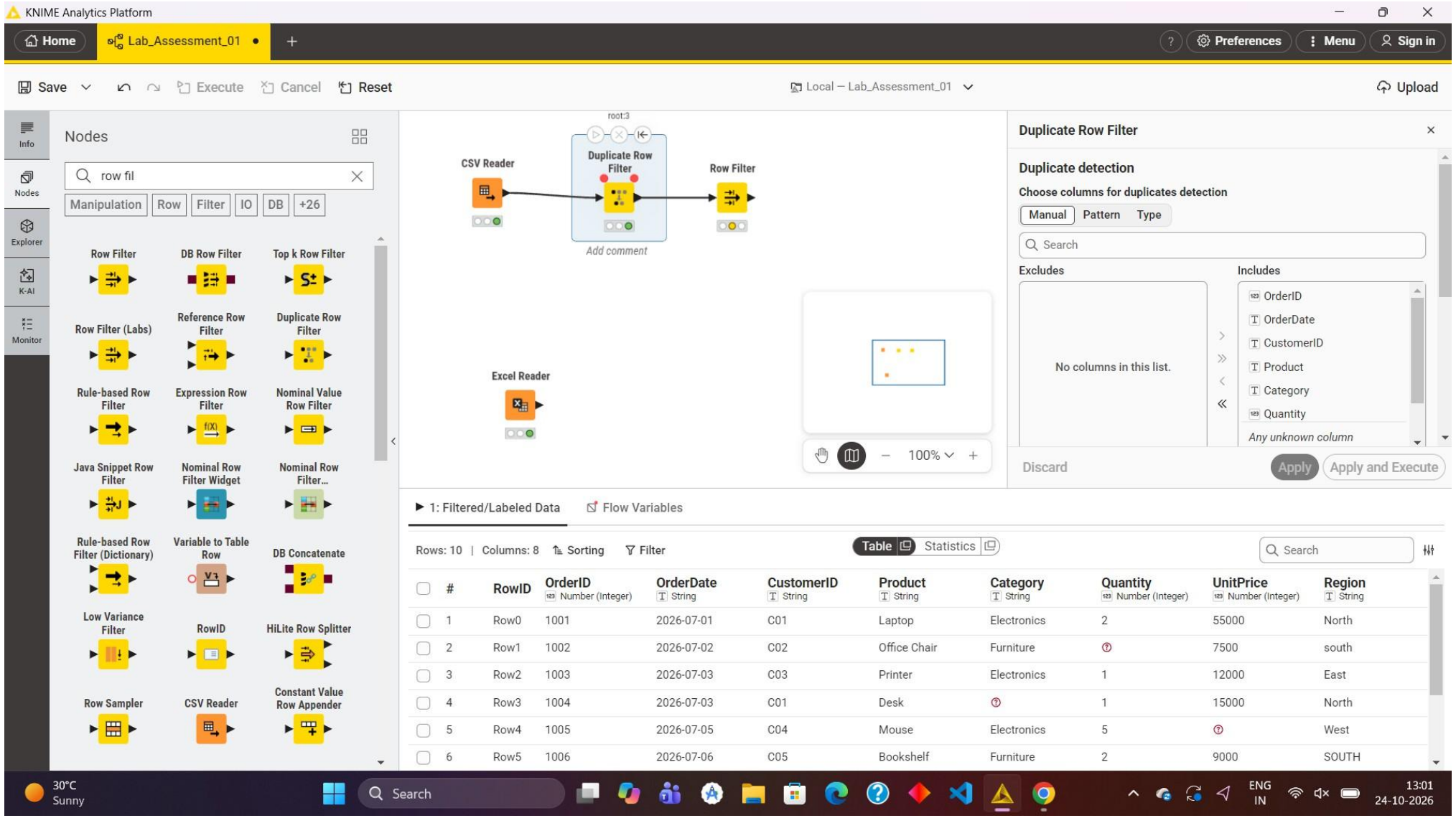
Importing the Customers Dataset



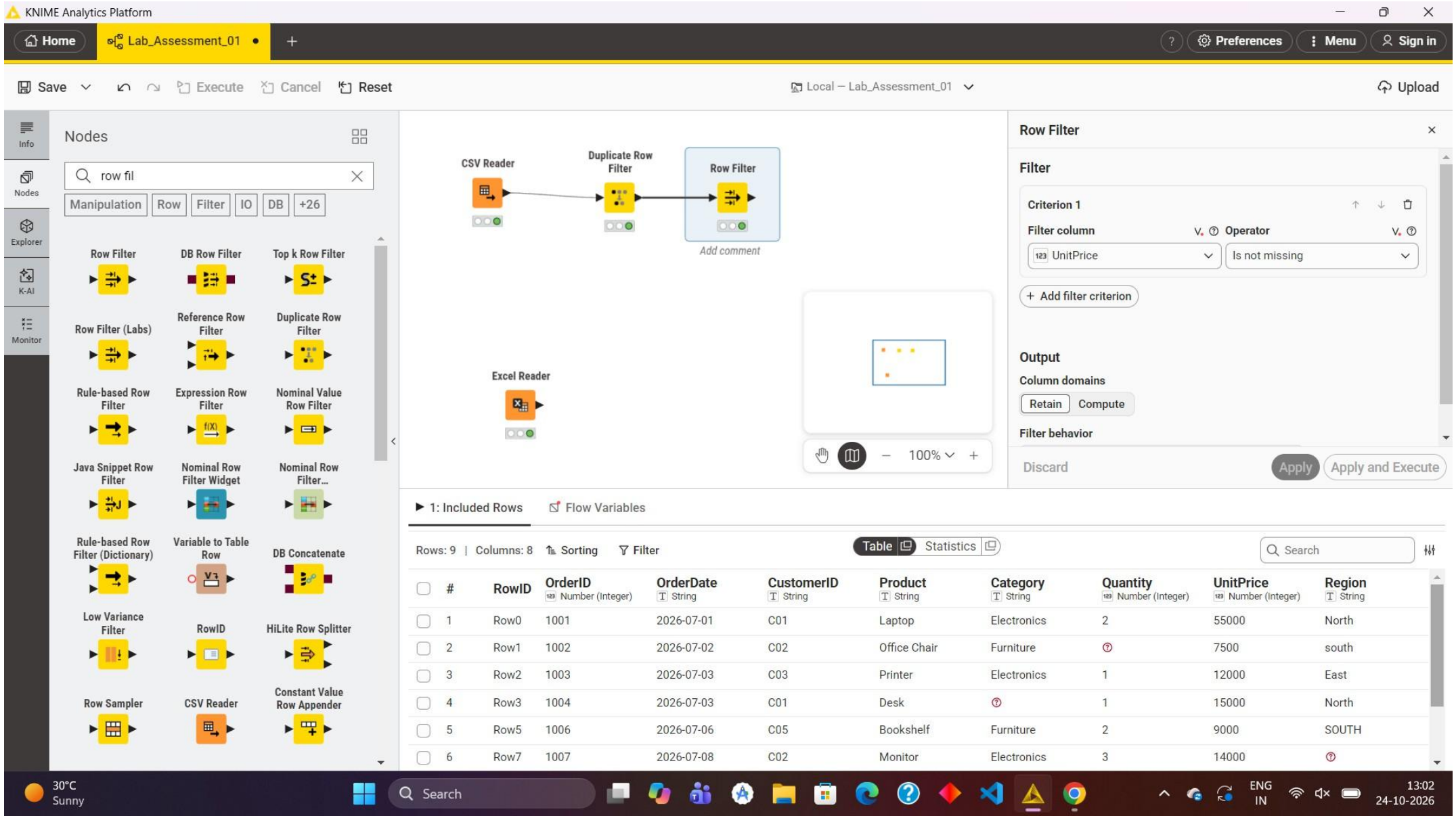
Building the Data-Cleaning Workflow



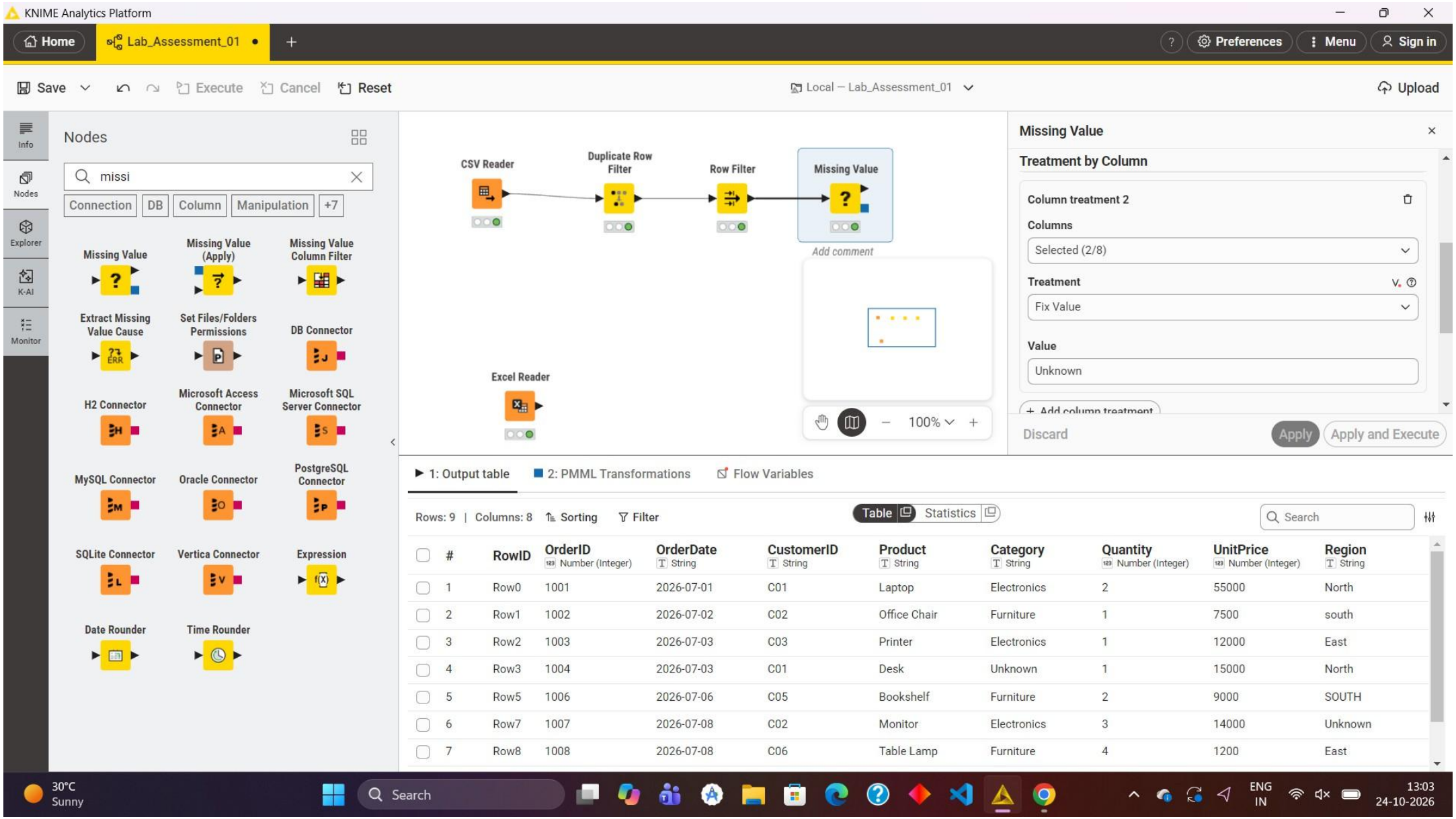
Removing the Duplicate Order



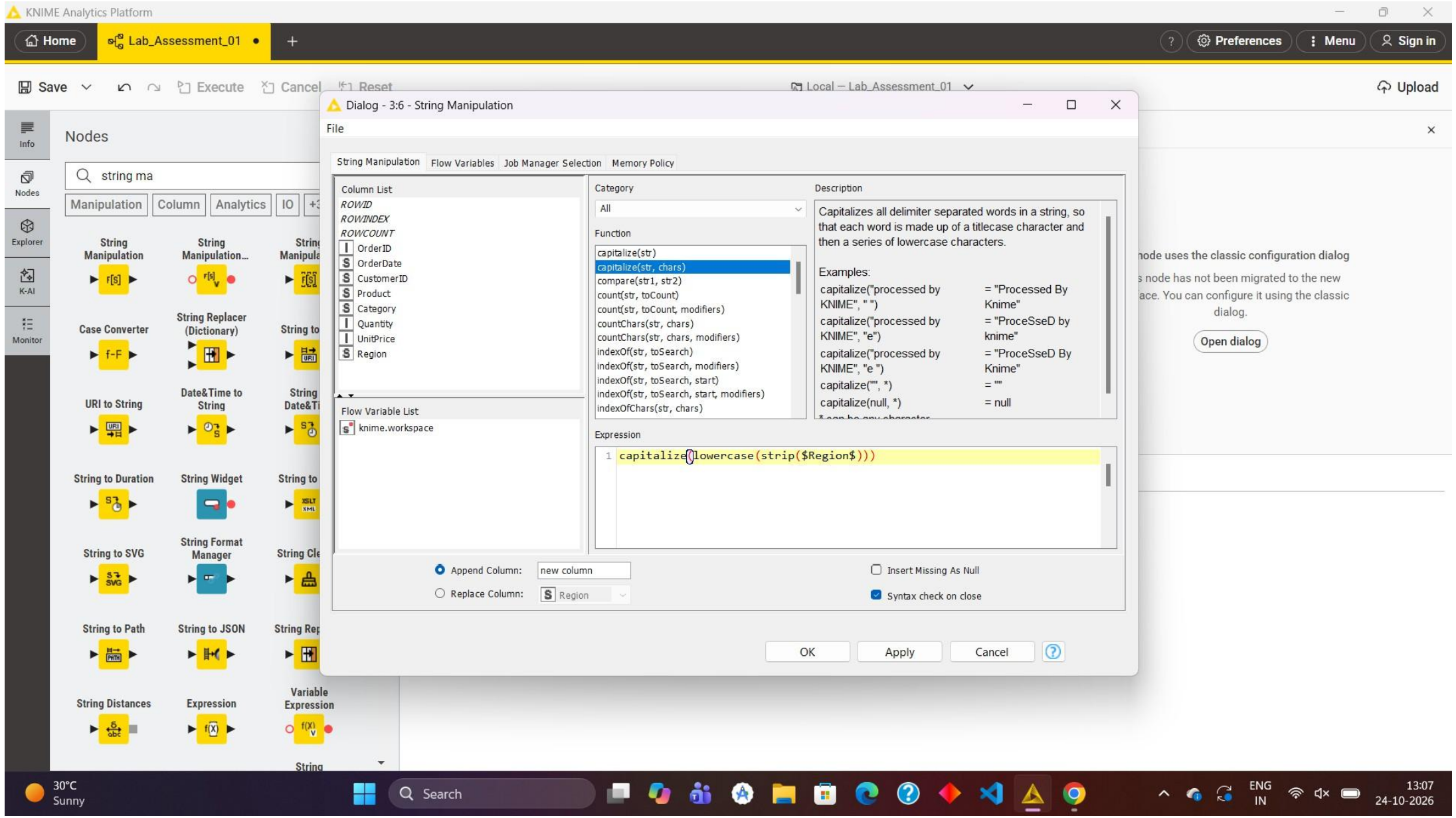
Filtering Rows with Missing UnitPrice



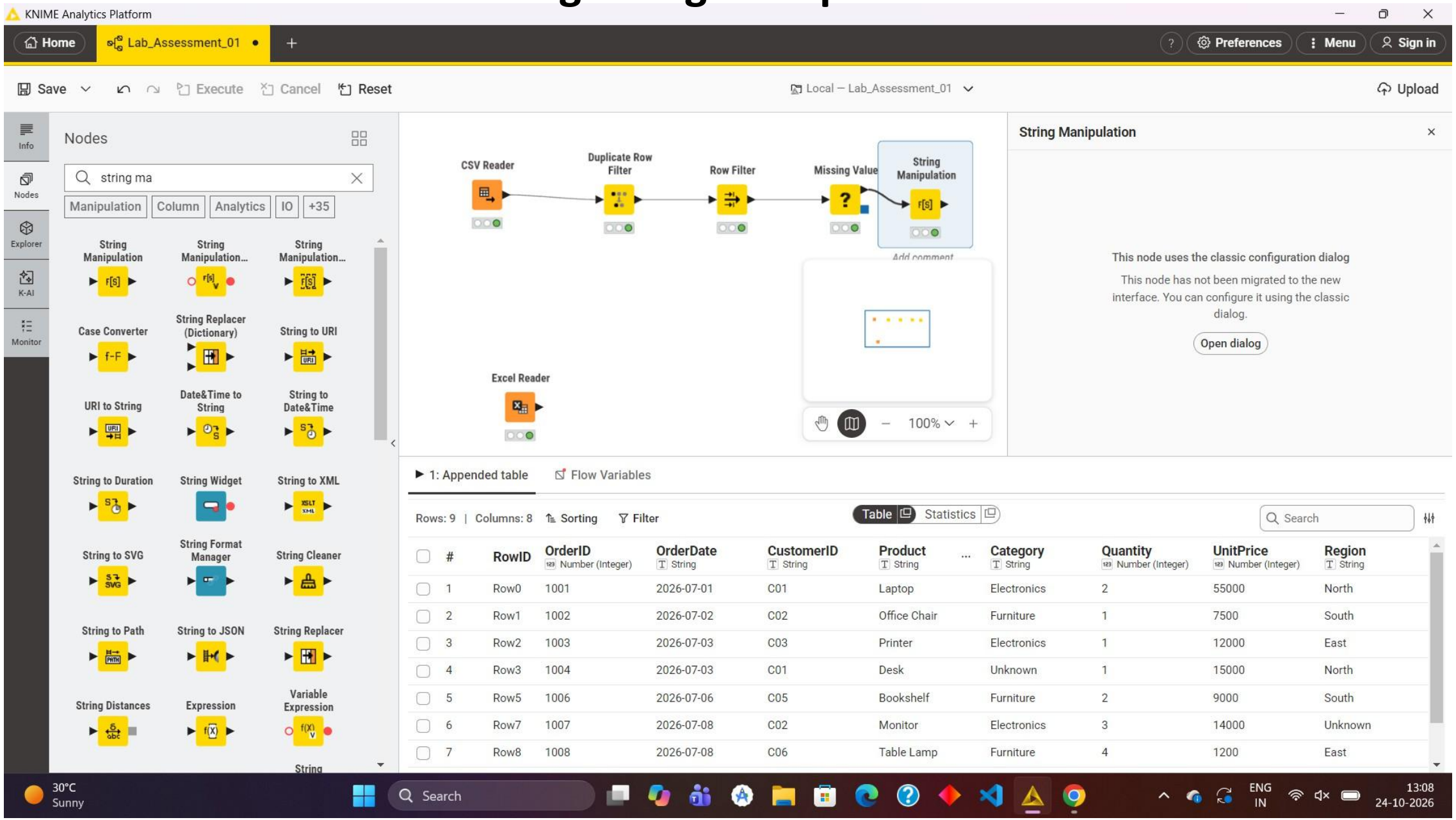
Configuring Missing-Value Handling



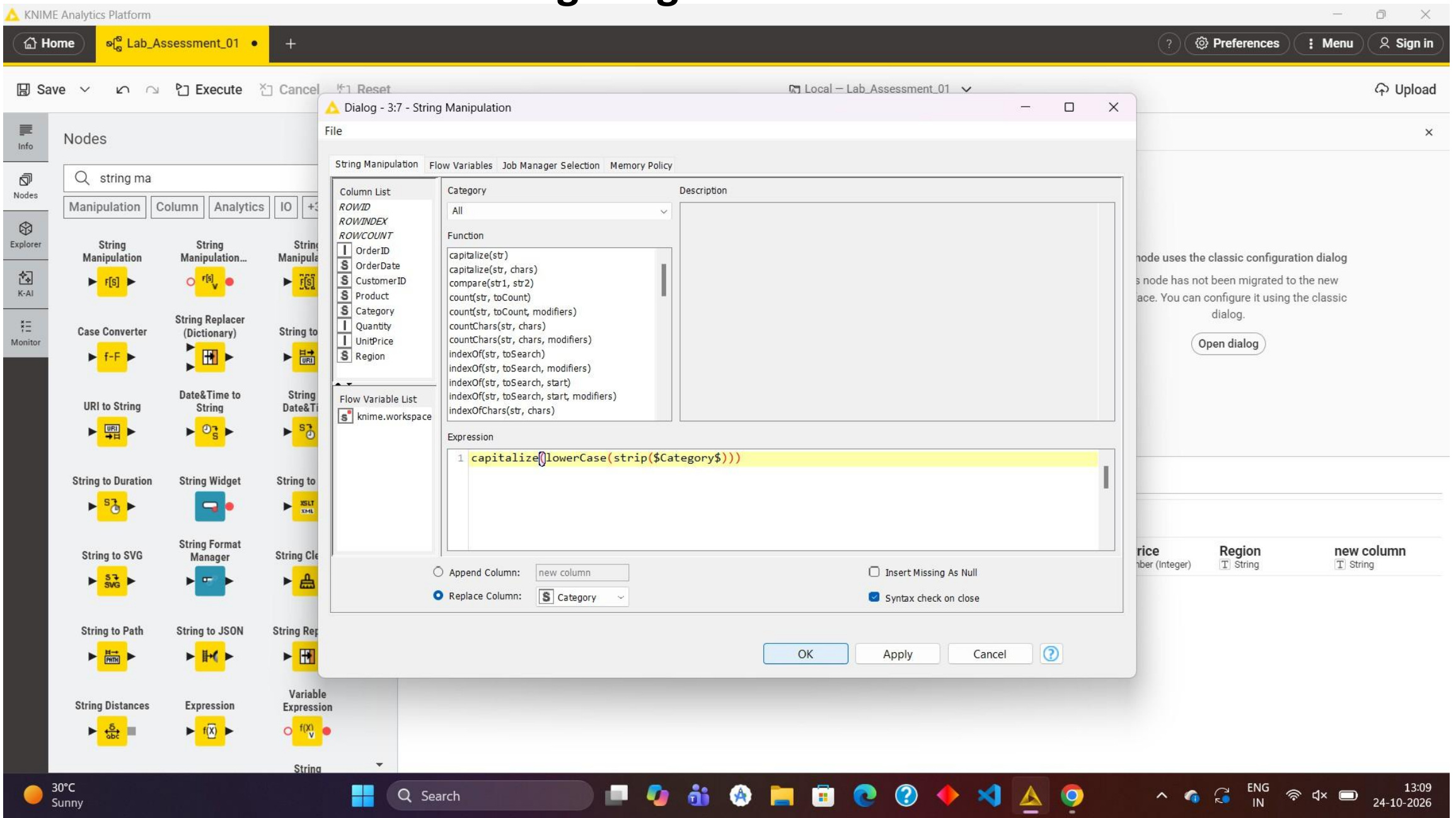
Setting Replacement Values



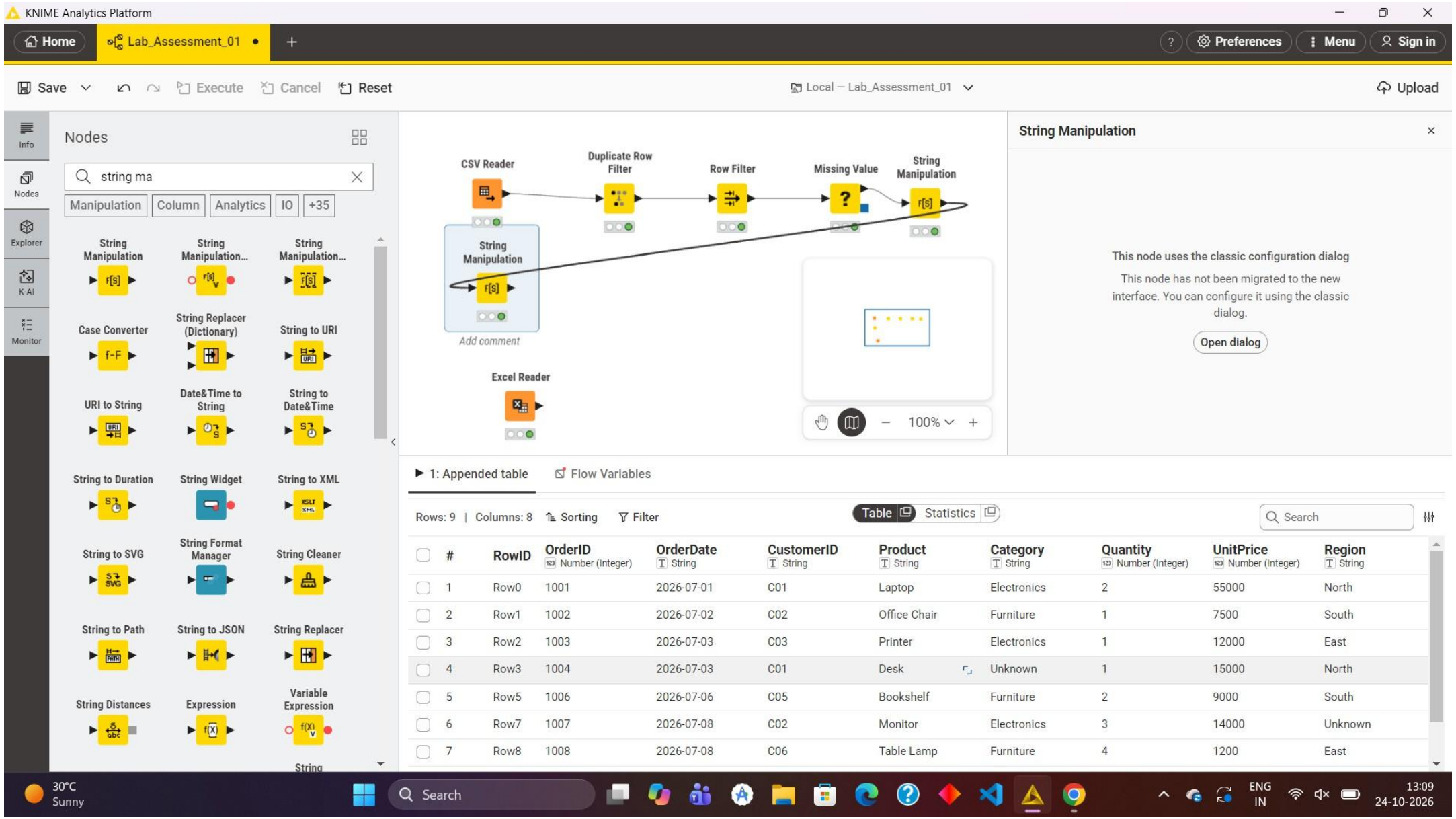
Adding String Manipulation



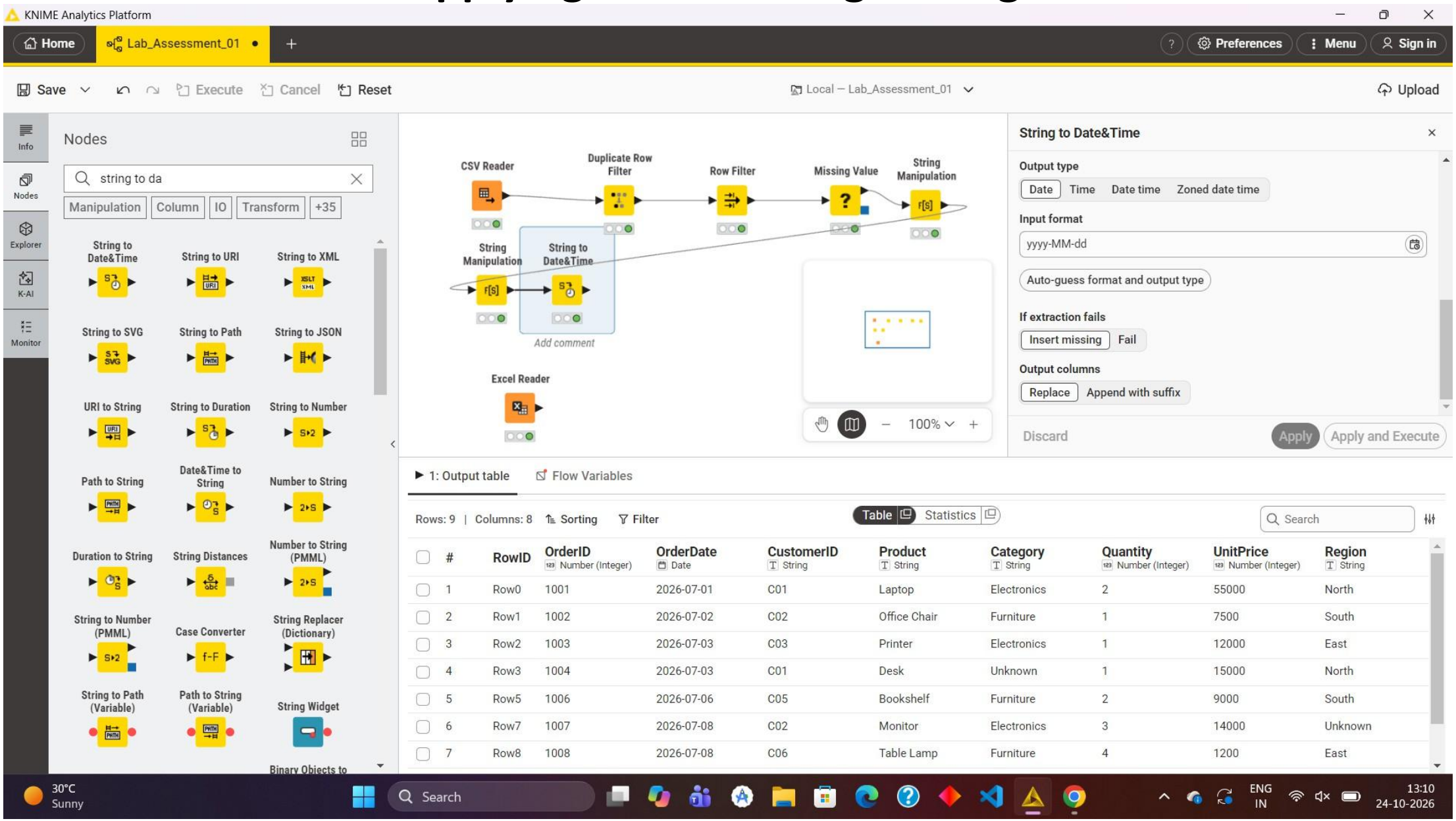
Configuring Text Standardization



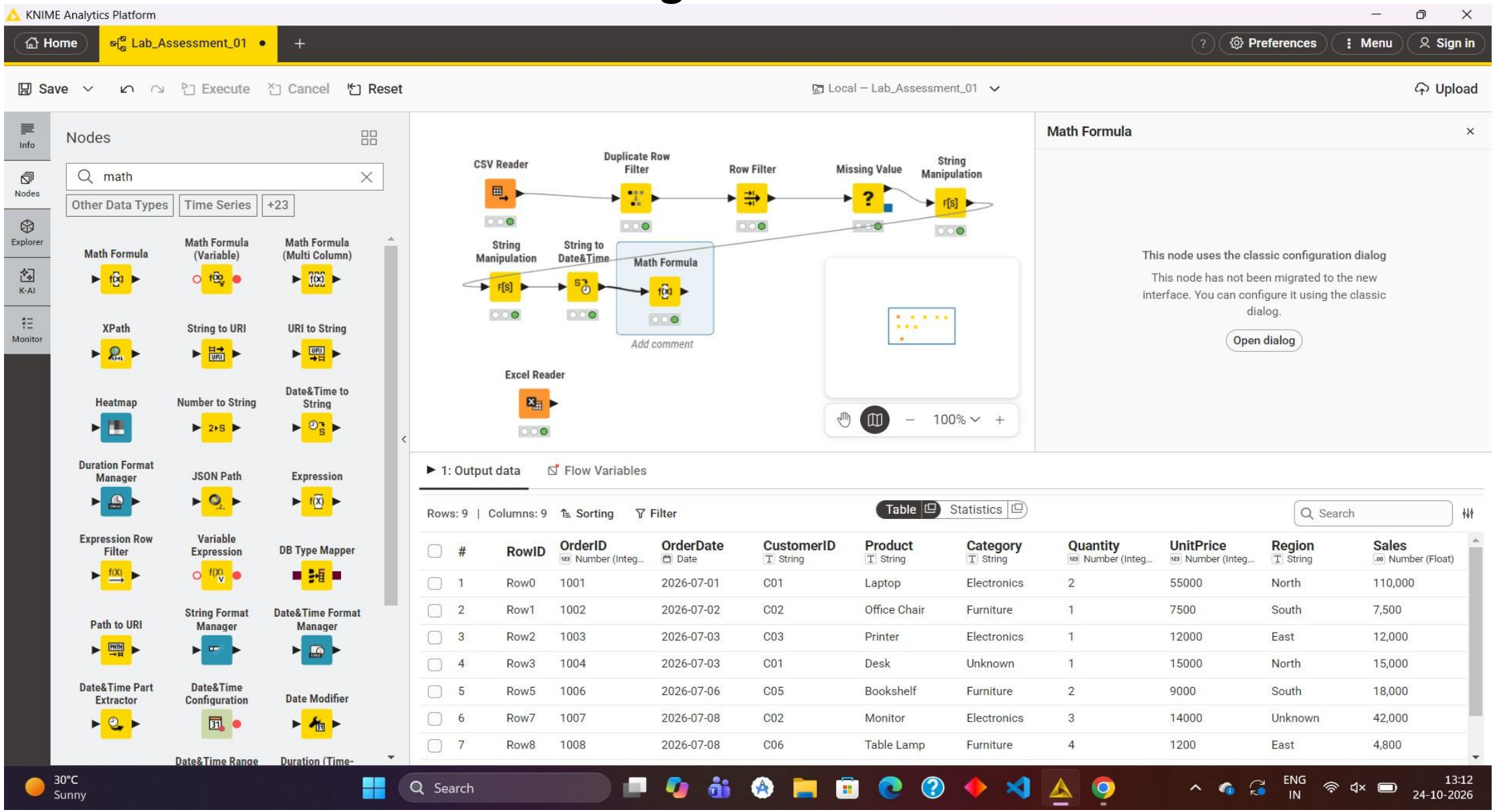
Checking the Cleaning Workflow



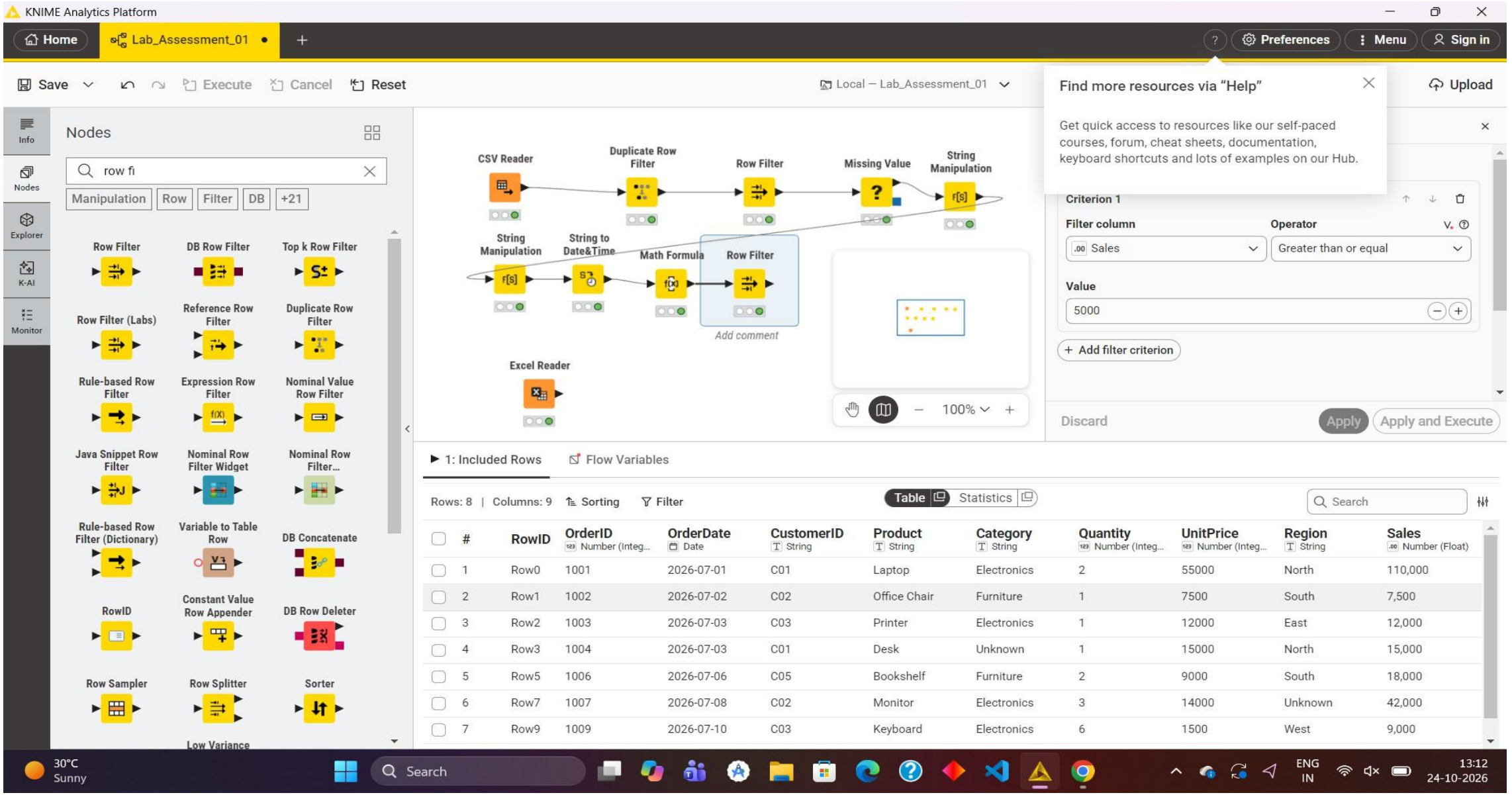
Applying Text-Cleaning Settings



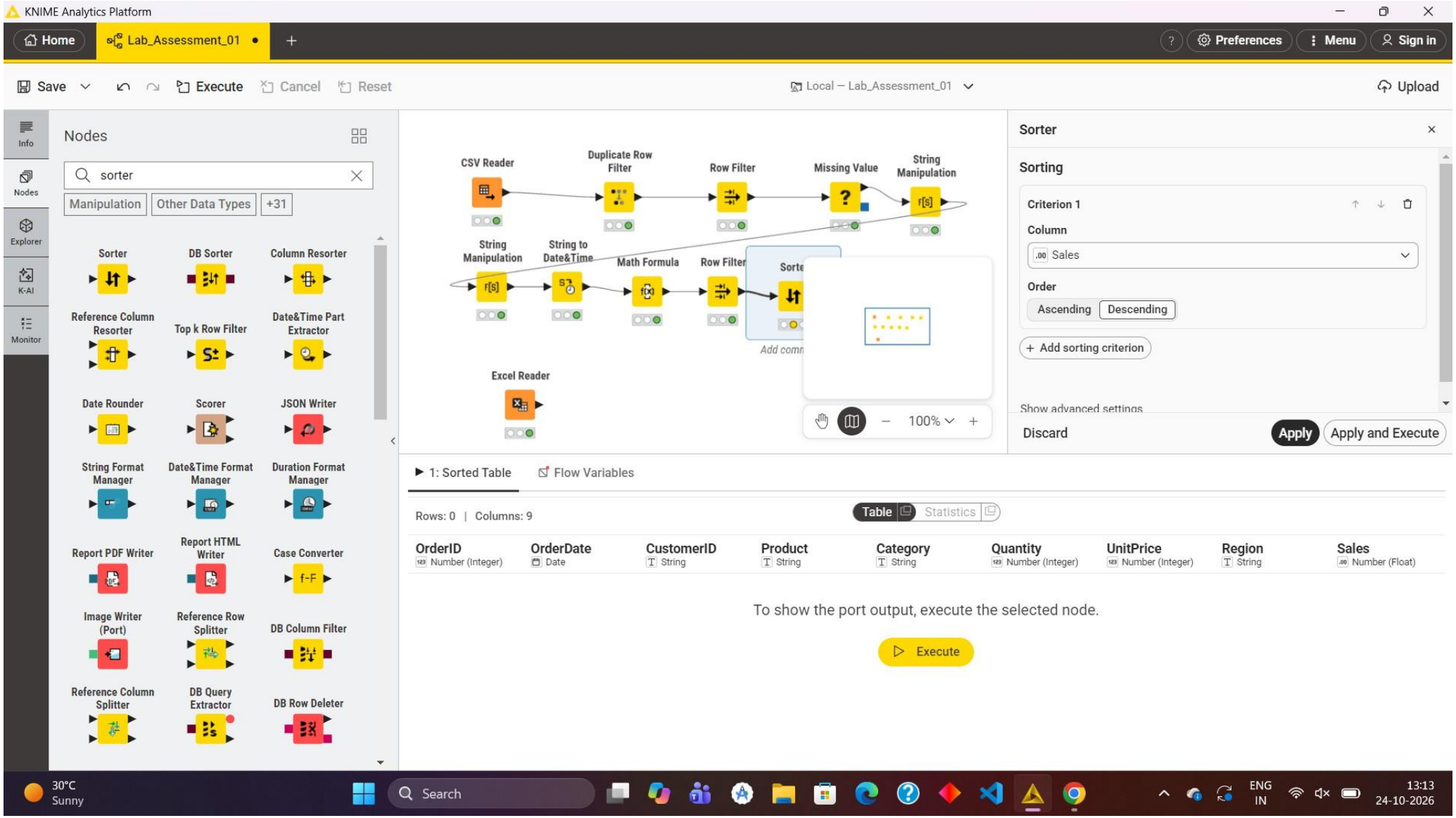
Creating the Sales Column



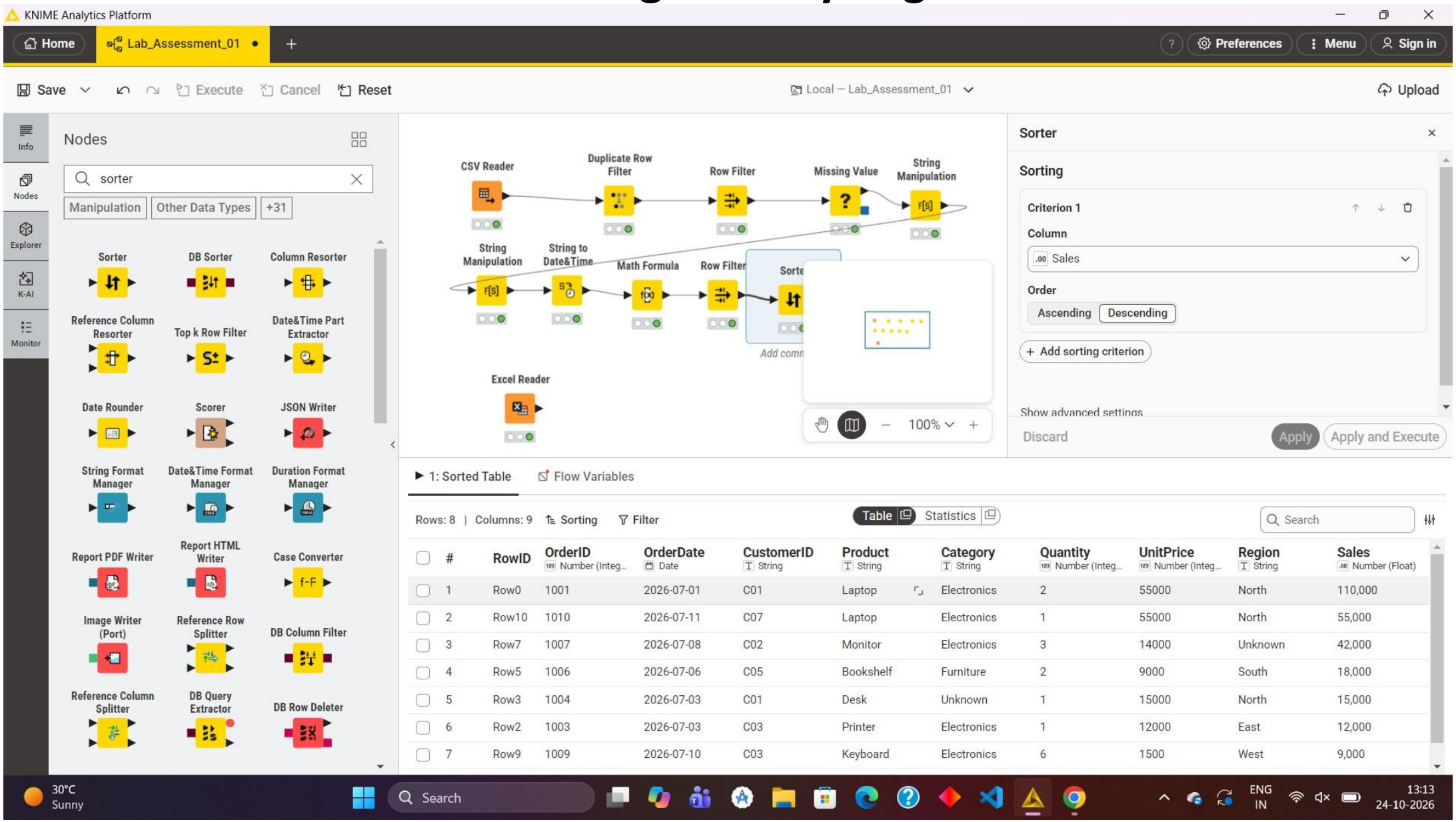
Filtering Orders by Sales



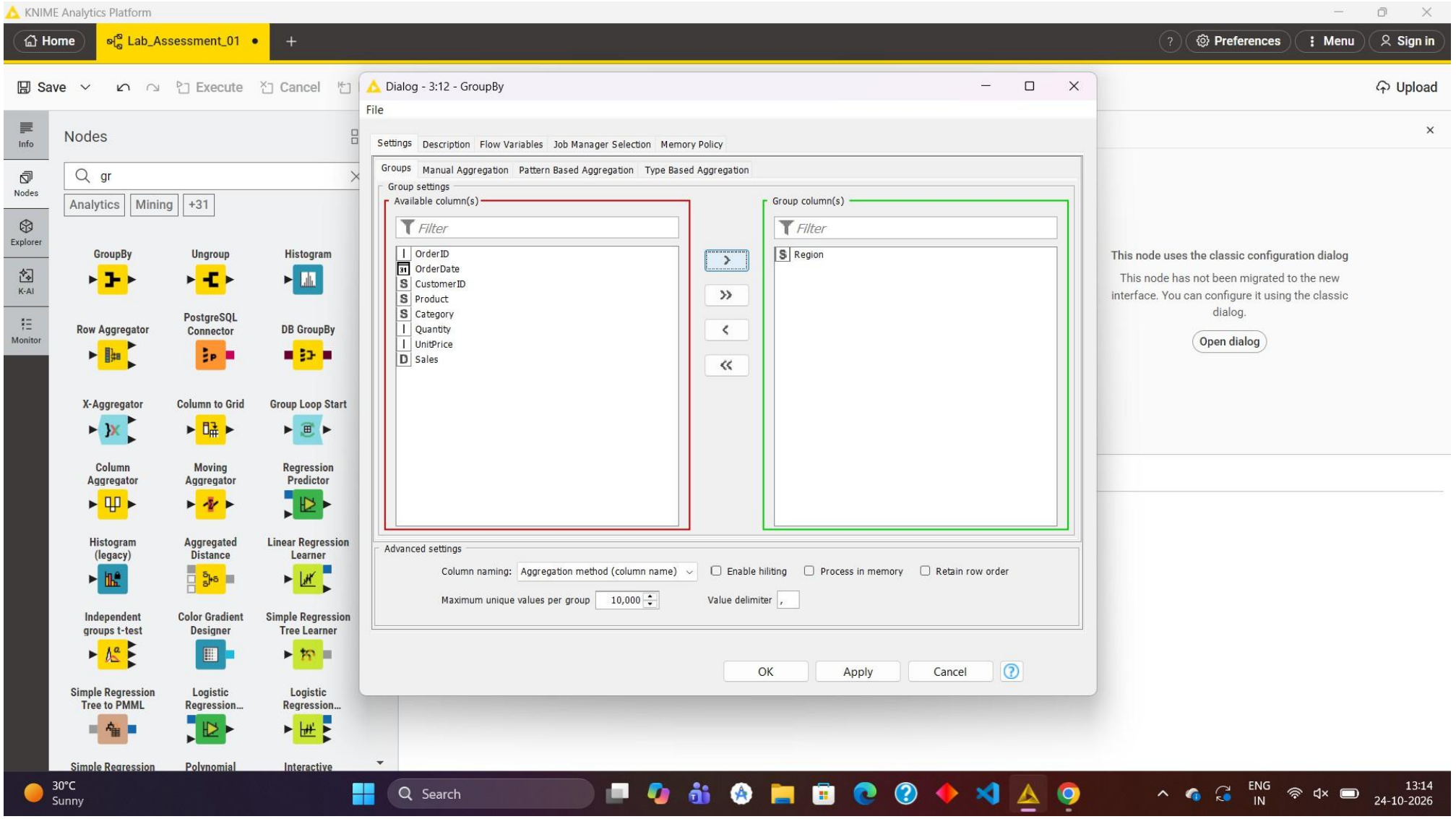
Sorting Sales from Highest to Lowest



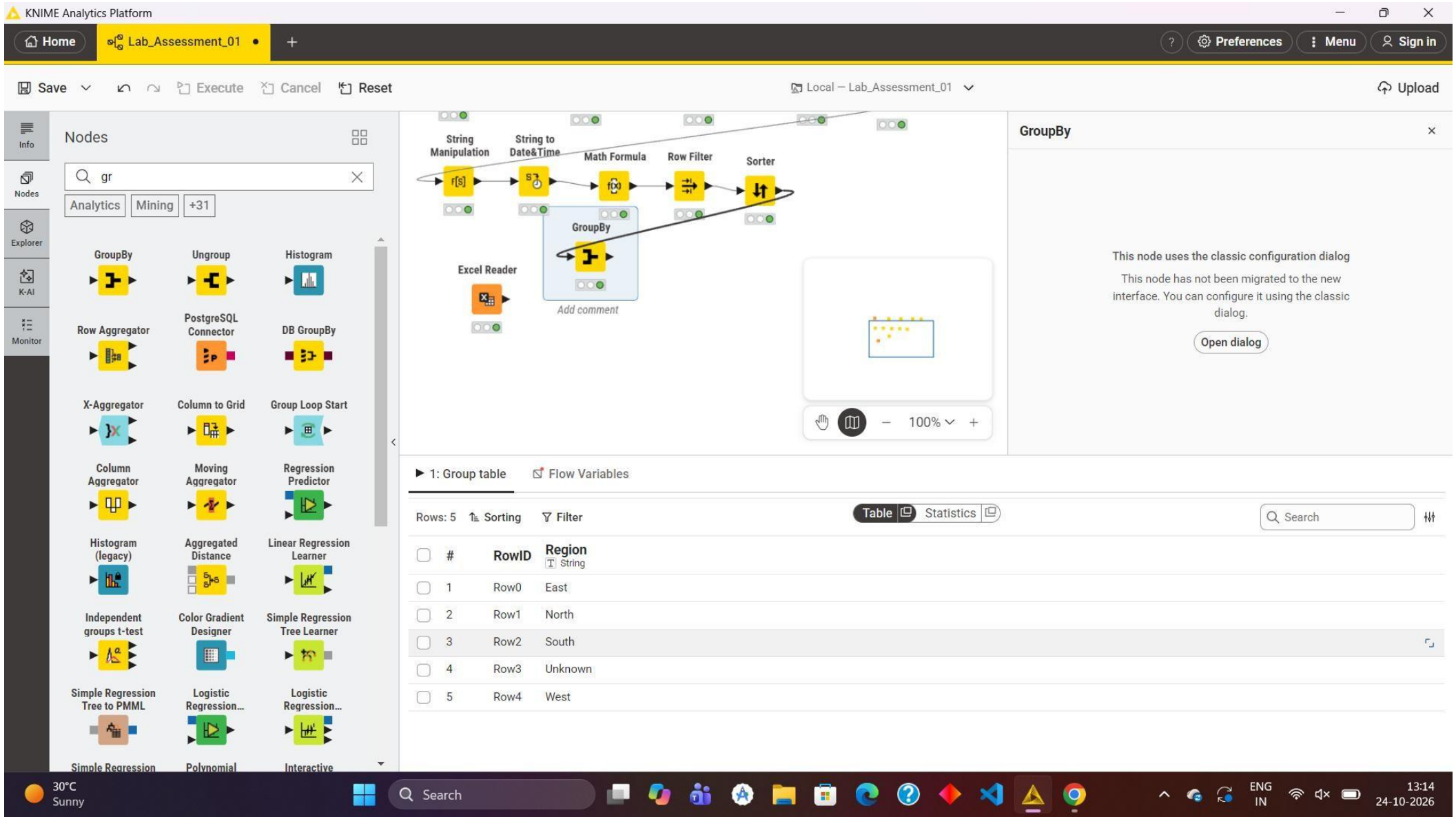
Summarizing Sales by Region



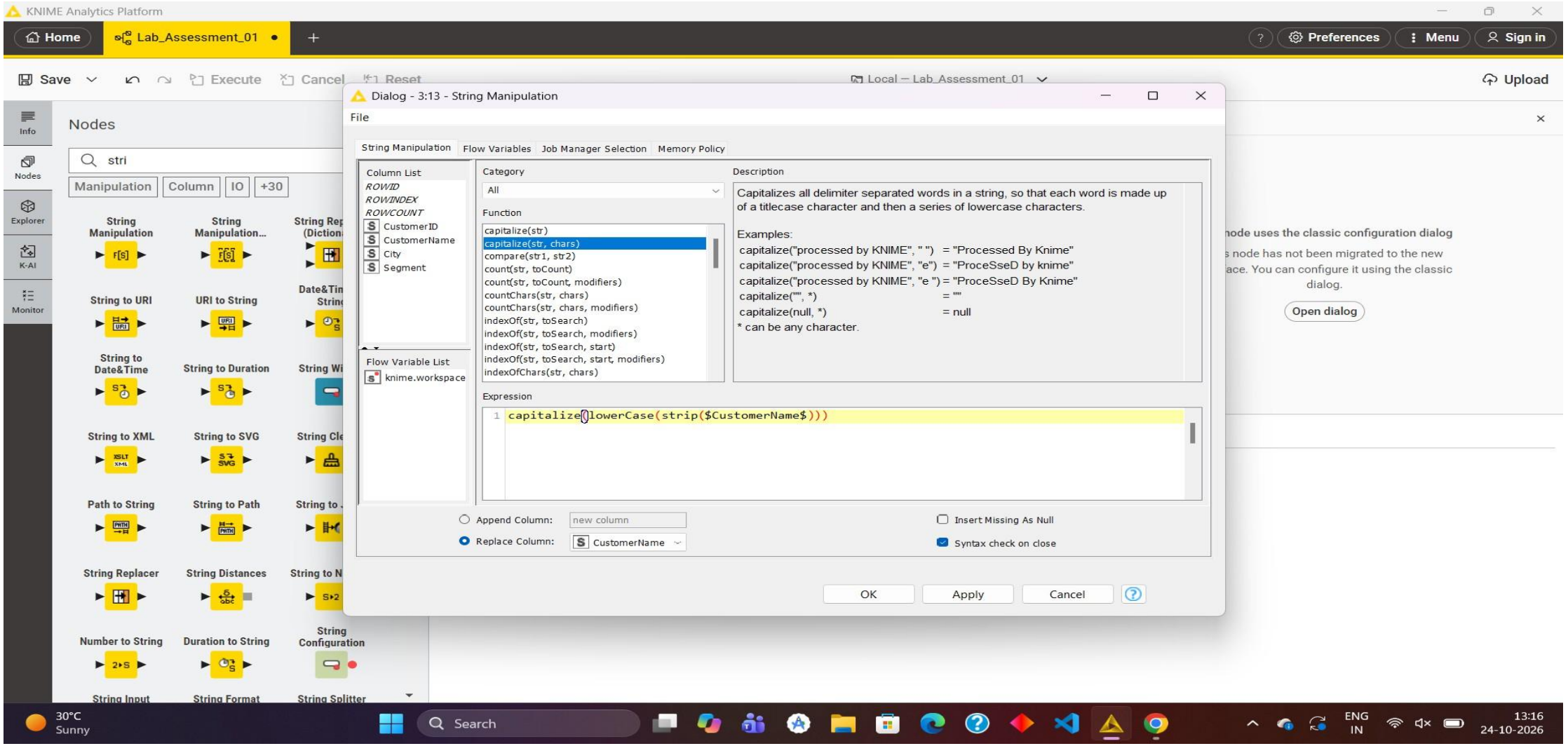
Preparing the Dataset Join



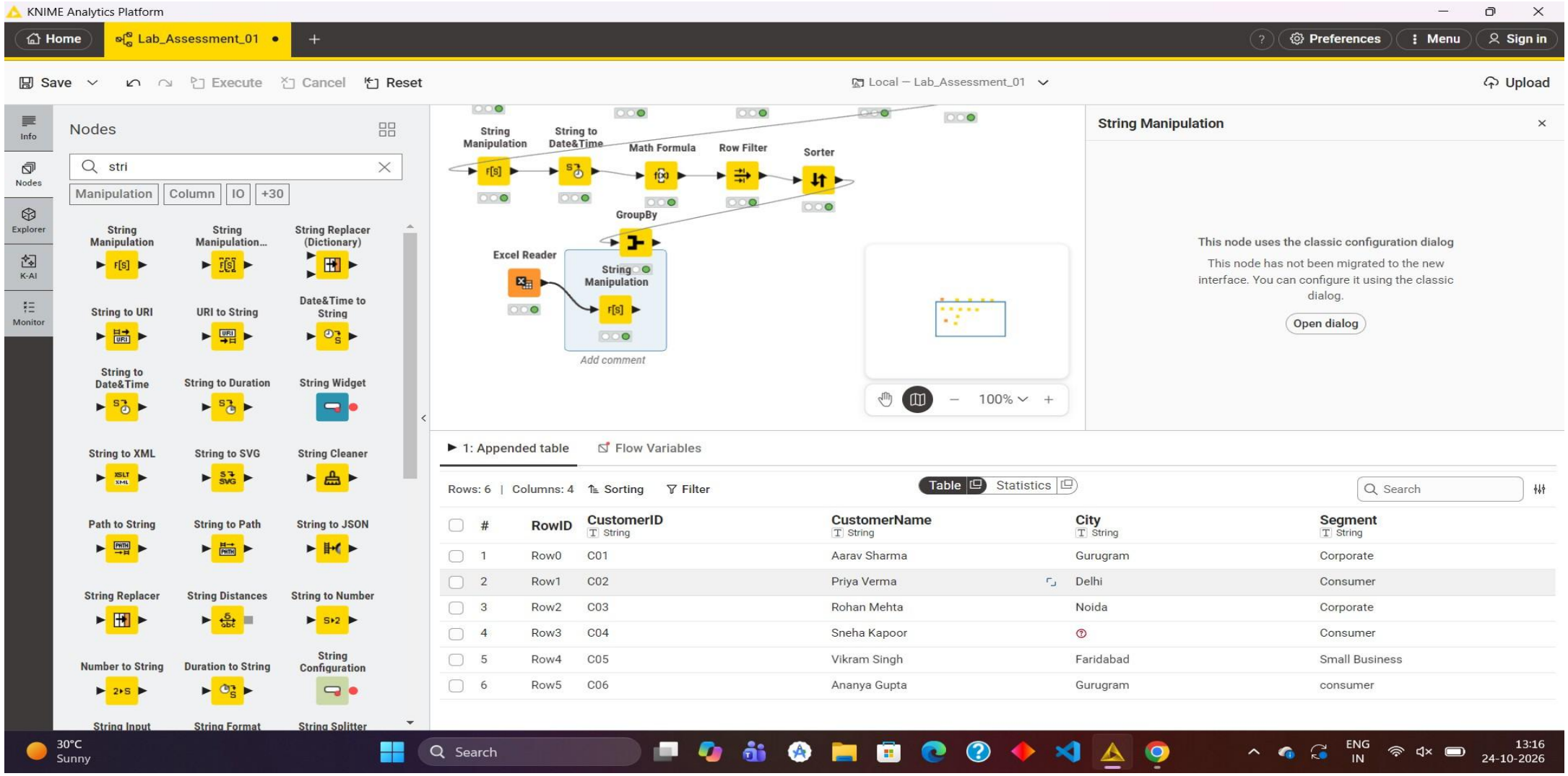
Configuring the Join



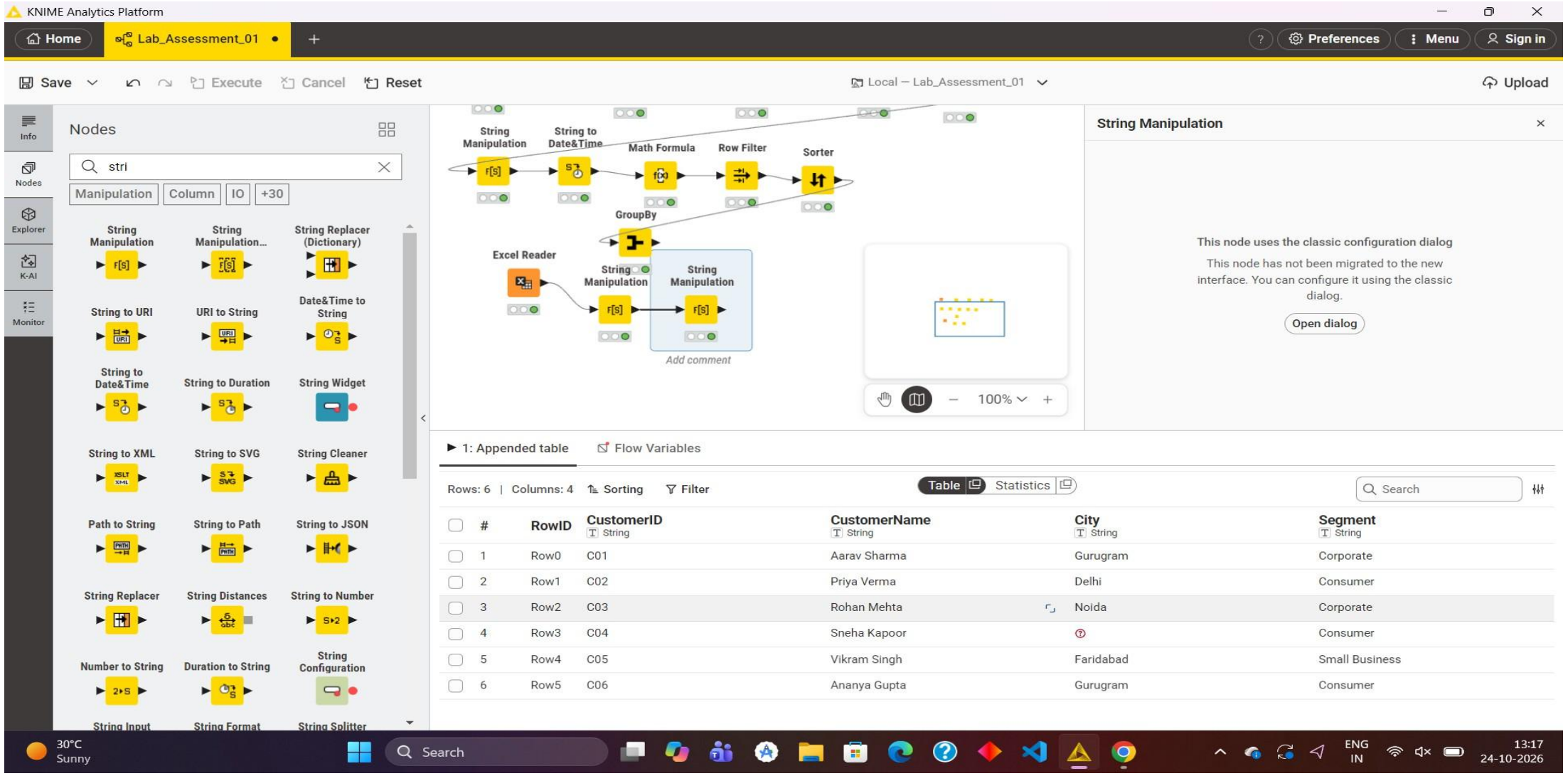
Reviewing the Joined Workflow



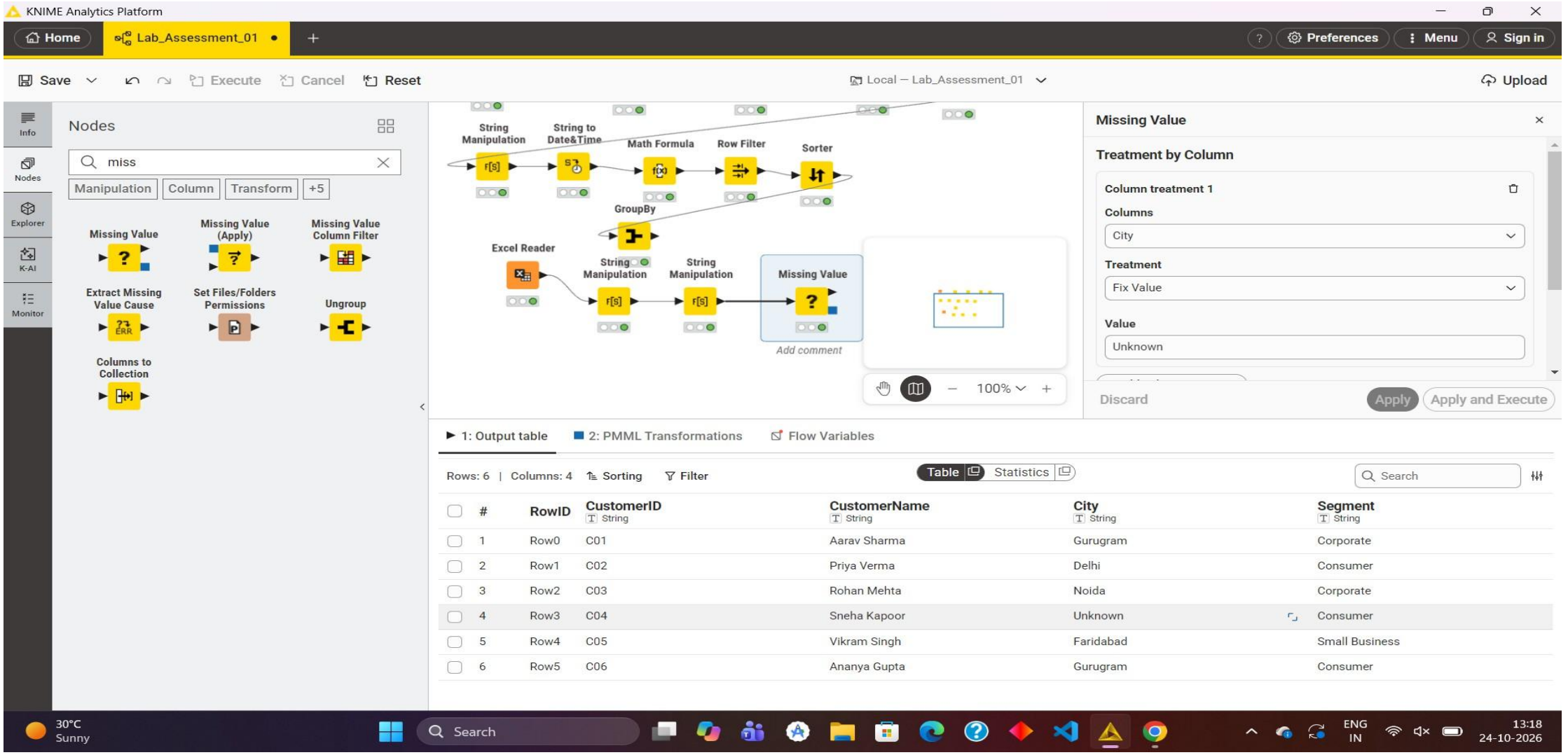
Preparing the Final Output



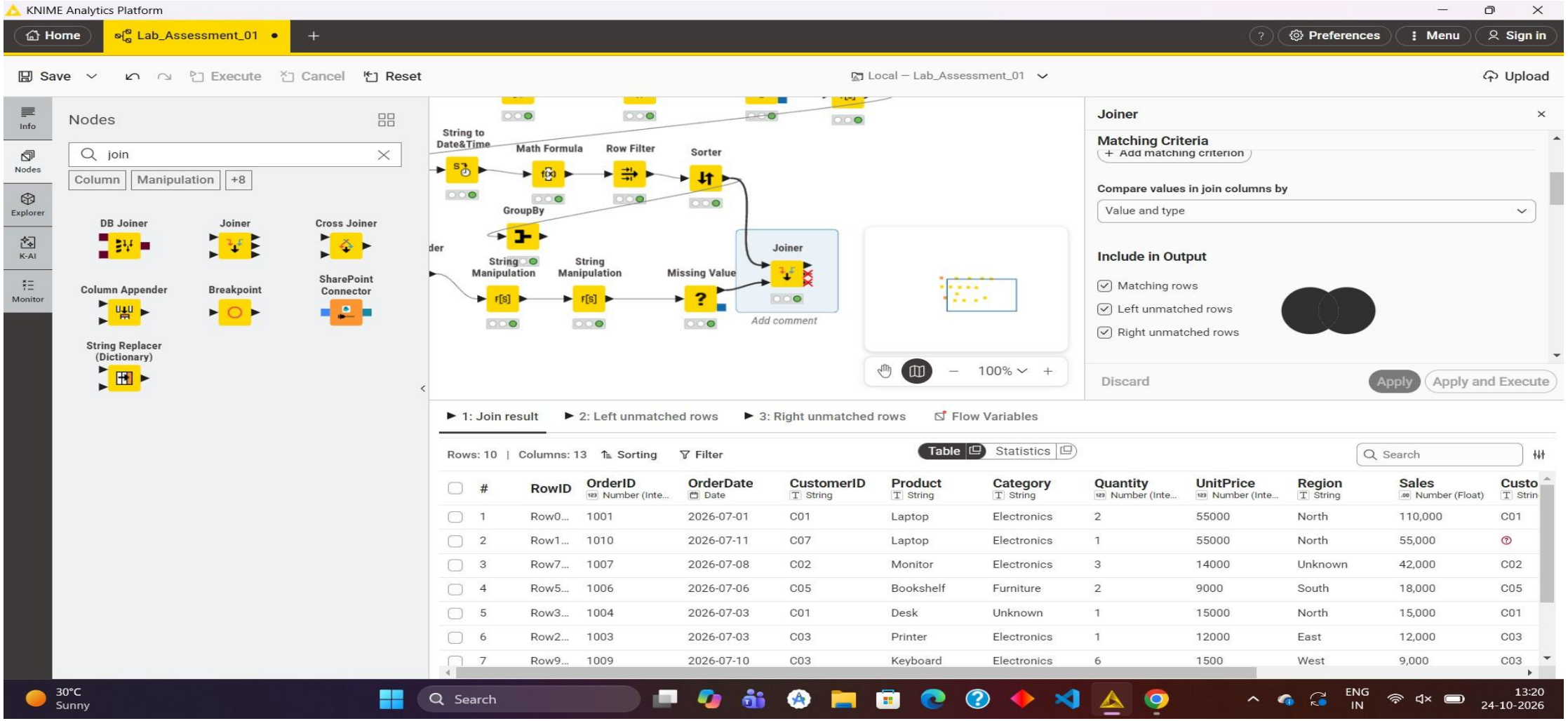
Reviewing the Final Workflow Structure



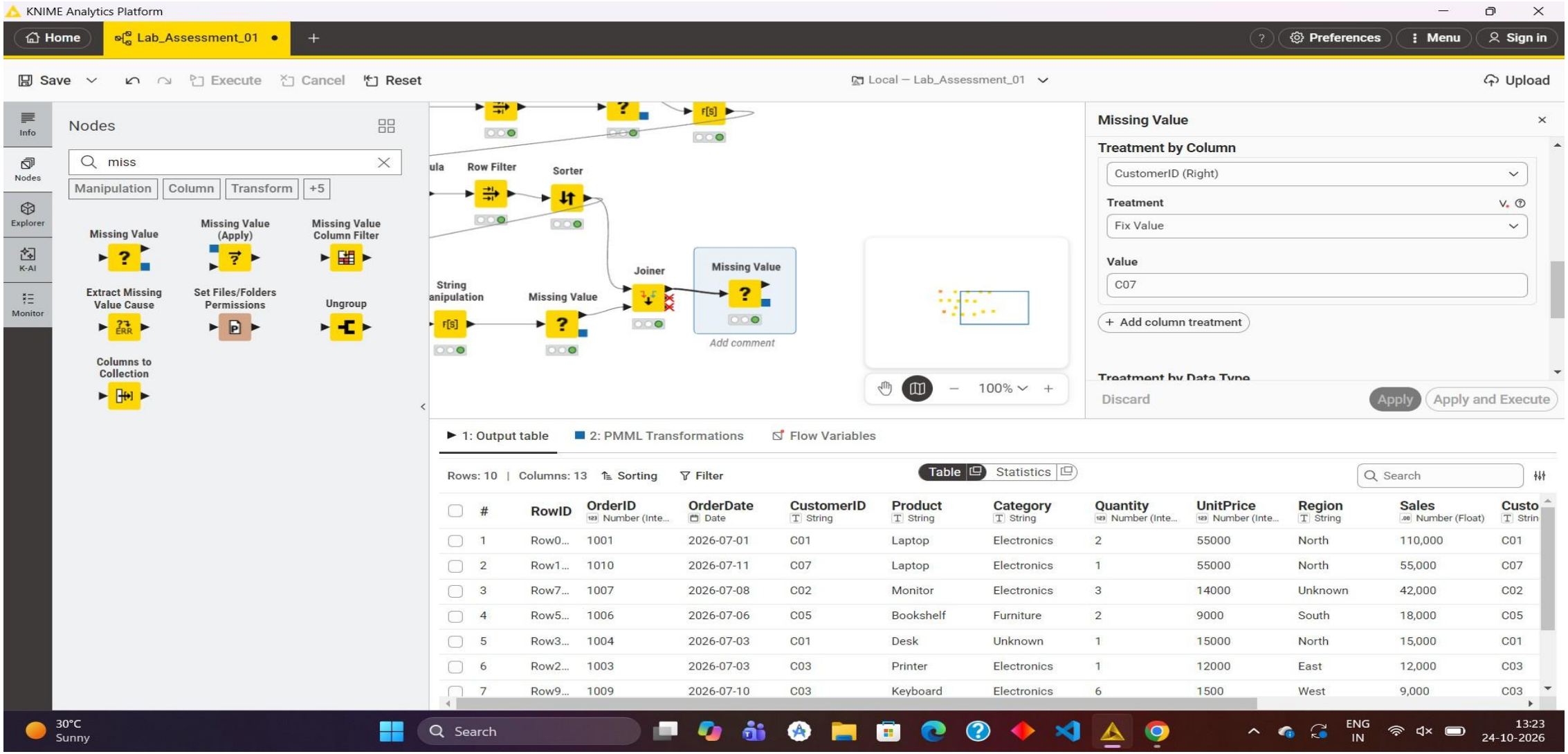
Checking the Joined Data Table



Verifying the Final Columns



Configuring the CSV Output



Final KNIME Workflow

